

Nonparametric Cyclic Registration and Distance-Based Clustering of Electrical Event Waveforms

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Abstract

We study unsupervised clustering of densely sampled electrical event waveforms observed with unknown cyclic offsets. Direct functional distances confound class-dependent morphology with nuisance phase, whereas unrestricted time warping can alter the duration and relative timing of fault signatures. We propose **TC-CyclicClust**, a nonparametric registration-aware procedure that smooths and centers each waveform, minimizes a squared-error dissimilarity over circular shifts with a local boundary correction, and applies hierarchical clustering to the resulting pairwise dissimilarities. Unlike model-based simultaneous registration and clustering, the procedure does not posit cluster-specific templates, Gaussian mixture components, random-effects distributions, or a linearization of the registration map. For the core cyclic-shift version, we establish a finite-sample exact-recovery guarantee under an additive signal-plus-noise model. The separation condition is expressed through the registered within-cluster diameter and between-cluster gap and makes explicit how rolling-window smoothing reduces stochastic fluctuations while potentially modifying signal separation. The method is motivated by high-frequency vibration, magnetic-field, and voltage records from a substation transformer, where event signatures are localized but their observed positions vary across recordings. In this setting, cyclic registration is scientifically appropriate, and the proposed distance-based procedure yields more accurate clustering than a model-based functional clustering comparator.

Keywords: functional data; curve registration; cyclic alignment; nonparametric clustering; electrical waveforms; smoothing; exact recovery.

1 Introduction

High-frequency monitoring systems increasingly record electrical equipment through voltage, vibration, magnetic-field, and related sensing modalities. Such measurements produce collections of densely sampled waveforms whose shapes contain information about switching operations, operating-state changes, and equipment faults. In practice, however, labels for these events are scarce, incomplete, or available only after costly expert review. An important exploratory task is therefore to cluster recorded events into groups with similar physical signatures. The resulting groups can support event taxonomy, prioritization of expert inspection, and the construction of subsequent diagnostic or predictive models.

The motivating data arise from a sensing system deployed on a substation transformer. Each record spans a common observation window and contains measurements from vibration, magnetic-field, and voltage sensors. The event-related portion of a waveform is often short relative to the full record, and its location can vary substantially from one record to another because the recording origin is arbitrary relative to the event and to the underlying electrical cycle. Moreover, the sensor modalities have markedly different waveform morphologies: vibration channels may exhibit similar pulse-like patterns, whereas magnetic-field and voltage channels can contain complementary changes in level, direction, and transient shape. These features make threshold rules or modality-specific templates difficult to transfer across sensors. They also make direct Euclidean or L^2 comparison unreliable, because two waveforms generated by the same event mechanism may appear far apart solely because their salient features occur at different cyclic offsets.

This is a functional-data problem with both amplitude and phase variation. The classical registration literature distinguishes vertical, or range, variation from horizontal, or domain, variation and aligns curves before comparing their amplitudes (Marron et al.,

2015, Ramsay and Li, 1998). A broad family of methods estimates smooth monotone transformations of time, including landmark registration, dynamic time warping, and non-parametric synchronization (Gervini and Gasser, 2004, Kneip and Gasser, 1992, Wang and Gasser, 1997). These methods are appropriate when individual processes genuinely evolve at different local rates. The physical nuisance in the present application is more restrictive. A change in the origin of a periodic record should translate the complete waveform around a circle, but it should not compress one part of a fault transient and expand another. Local duration, rise and decay rates, and the spacing between peaks can themselves distinguish event types. An unrestricted monotone warp can therefore remove scientifically meaningful morphology and may align waveforms from different event classes too aggressively. This concern is consistent with the well-known nonidentifiability and distortion issues that arise when registration maps are overly flexible. We consequently treat phase as a cyclic translation and compare waveforms on the quotient space induced by the finite group of circular shifts. Cyclic registration removes the arbitrary origin while preserving all relative temporal distances within the waveform.

Clustering functional observations has been approached from both distance-based and model-based perspectives. Distance-based procedures include functional variants of k -means, clustering after transformation and smoothing, and robust or center-based methods (Jacques and Preda, 2014, Serban and Wasserman, 2005, Tarpey and Kinateder, 2003). In the time-series literature, k -Shape constructs a shift-invariant similarity from normalized cross-correlation (Paparrizos and Gravano, 2015). Model-based procedures instead represent cluster-specific curves through basis expansions or random-effects models and assign observations through a mixture likelihood; examples include clustering sparsely observed curves and mixed-effects spline models (Chiou and Li, 2007, James and Sugar, 2003). Registration is often performed as a preliminary step, although separating registration from

clustering can be problematic because the appropriate alignment depends on the unknown cluster structure.

A particularly relevant comparator is the simultaneous aligning and clustering k -centres procedure of Liu and Yang (2009). Their SACK model gives a coherent probabilistic treatment of phase and amplitude variation: cluster means are represented by B-splines, individual time and level shifts are modeled as Gaussian random effects, and cluster membership and latent shifts are estimated with an EM-type algorithm. A first-order Taylor expansion moves the time shift into the response scale and yields tractable updates. This construction has important advantages when likelihood-based uncertainty quantification, cluster-specific mean curves, or irregular sampling are central. It also imposes a shape-invariant mixture model, distributional assumptions on the random effects and errors, a spline representation, and an approximation whose adequacy depends on the magnitude of the time shifts. Sharp, localized, densely sampled electrical waveforms need not conform well to these assumptions. In particular, a cluster may contain a family of related fault signatures without admitting a single smooth template and Gaussian perturbation model.

We develop a nonparametric alternative in which registration is built directly into the pairwise dissimilarity. Each waveform is first smoothed by a rolling-window operator and centered to remove an irrelevant vertical level. For each pair, squared-error discrepancy is minimized over a grid of circular shifts; the implemented version also uses a local linear correction to reduce an artificial jump at the wrap boundary. The resulting registration-adjusted dissimilarity matrix is supplied to hierarchical clustering. Thus, registration is not a one-time preprocessing operation relative to a global template. It is recomputed within every comparison that defines the geometry seen by the clustering algorithm. The procedure is therefore joint at the level of the clustering criterion, while remaining fully distance based and avoiding a likelihood for the waveform population.

The statistical formulation is deliberately more general than a prototype model. We do not assume that all members of a cluster are shifted copies of one mean curve. Instead, clusters are characterized by a registered separation condition: after optimizing over cyclic shifts, the largest relevant within-cluster discrepancy is smaller than the smallest between-cluster discrepancy. Under an additive signal-plus-Gaussian-noise model, we establish a finite-sample condition under which hierarchical clustering exactly recovers the true partition, up to a permutation of labels. The bound makes explicit the roles of registration and smoothing. Registration changes the deterministic within- and between-cluster geometry, whereas the rolling window transforms white noise into correlated Gaussian noise and reduces the leading pure-noise fluctuation. Smoothing can therefore improve recovery by lowering stochastic variability, but it can also attenuate features that contribute to cluster separation; the theory describes this trade-off through the separation of the smoothed, registered signals.

The proposed approach is evaluated on electrical event waveforms from the transformer sensing study and against both unregistered or pattern-based procedures and the model-based SACK method. The empirical results show that cyclic registration is essential when event location varies and that the distance-based method is better suited to the sharply localized and heterogeneous waveform shapes in this application. In particular, the proposed method gives more accurate clustering than the model-based comparator while retaining a transparent interpretation: two records are close when one can be moved around the cyclic domain, without nonlinear deformation, to match the other.

The main contributions are threefold.

1. We introduce an application-specific registration principle for electrical waveforms.

The nuisance transformation is a cyclic shift induced by periodic operation and ar-

bitrary record origin. Unlike general time warping, this transformation preserves event duration, local slopes, and relative peak locations, which are potentially fault-discriminative features.

2. We couple registration with nonparametric, distance-based clustering. Pairwise alignment is optimized inside the dissimilarity used by hierarchical clustering, without estimating a global template or specifying a finite-mixture, spline, random-effects, or Gaussian shape model. This provides a direct counterpart to model-based simultaneous registration and clustering and is particularly natural for densely sampled waveforms.
3. We quantify how registration and rolling-window smoothing affect cluster recovery. Our nonasymptotic analysis gives an exact-recovery condition in terms of the registered within-cluster diameter, the registered between-cluster separation, the noise level, and the smoothing window, thereby separating the deterministic and stochastic consequences of preprocessing.

The remainder of the paper is organized as follows. Section 2 introduces the trend-corrected cyclic dissimilarity and the hierarchical clustering algorithm. Section 3 presents the signal-plus-noise model and the exact-recovery results. **Section 4 gives the proofs.** Subsequent sections describe the transformer waveform data, simulation experiments, comparisons with competing methods, and practical guidance for choosing the smoothing window and shift grid.

2 Algorithm: Distance-Based Clustering of Waveforms

For $i = 1, \dots, n$, let $\mathbf{x}_i = (\mathbf{x}_i^{(1)}, \dots, \mathbf{x}_i^{(p)}) \in \mathbb{R}^{T \times p}$, where, $\mathbf{x}_i^{(r)} = (x_{i1}^{(r)}, \dots, x_{iT}^{(r)})^\top \in \mathbb{R}^T$, $r = 1, \dots, p$, denote the observed p -dimensional waveform measured at T equally spaced time points. The observed dataset is $\mathbb{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$. Let $A_\omega : \mathbb{R}^T \rightarrow \mathbb{R}^\tau$ denote the rolling-average operator with window length ω , where $\omega \geq 1$ and $\tau = T - \omega + 1$. For any $\mathbf{x} \in \mathbb{R}^T$,

$$(A_\omega \mathbf{x})_t = \frac{1}{\omega} \sum_{j=0}^{\omega-1} x_{t+j}, \quad t = 1, \dots, \tau.$$

We apply A_ω separately to each coordinate and define the smoothed waveform $\mathbf{y}_i = A_\omega \mathbf{x}_i = (A_\omega \mathbf{x}_i^{(1)}, \dots, A_\omega \mathbf{x}_i^{(p)}) \in \mathbb{R}^{\tau \times p}$, $i = 1, \dots, n$.

To remove differences in vertical level, we subsequently center each smoothed coordinate by replacing $\mathbf{y}_i$ with $\tilde{\mathbf{y}}_i = \mathbf{y}_i - \mathbf{1}_\tau \bar{\mathbf{y}}_i^\top$, where, $\bar{\mathbf{y}}_i = \tau^{-1} \mathbf{y}_i^\top \mathbf{1}_\tau \in \mathbb{R}^p$. Thus, $\tilde{\mathbf{y}}_i = P_\tau A_\omega \mathbf{x}_i$, where, $P_\tau = I_\tau - \tau^{-1} \mathbf{1}_\tau \mathbf{1}_\tau^\top$, and both A_ω and P_τ act separately on the columns of $\mathbf{x}_i$.

To compare two centered waveforms while allowing for temporal misalignment, we use a shift-adjusted dissimilarity. For an integer r , define cyclic reduction $[r]_\tau = 1 + \{(r-1) \bmod \tau\}$. For $h \in \{0, \dots, \tau-1\}$, let $S_h : \mathbb{R}^\tau \rightarrow \mathbb{R}^\tau$ denote the circular right-shift operator $(S_h \mathbf{v})_t = v_{[t-h]_\tau}$, for $t = 1, \dots, \tau$. Thus, if $\mathbf{v} = (v_1, \dots, v_\tau)^\top$, then $S_h \mathbf{v} = (v_{\tau-h+1}, \dots, v_\tau, v_1, \dots, v_{\tau-h})^\top$, with the convention that $S_0 \mathbf{v} = \mathbf{v}$. We refer to $\text{Wr}(\mathbf{v}; h) = (v_{\tau-h+1}, \dots, v_\tau)^\top$ as the wrapped segment and $\text{Le}(\mathbf{v}; h) = (v_1, \dots, v_{\tau-h})^\top$ as the leading segment, so that $S_h \mathbf{v} = (\text{Wr}(\mathbf{v}; h)^\top, \text{Le}(\mathbf{v}; h)^\top)^\top$.

Pure circular shifting can introduce artificial discontinuities at the wrap-around boundary. To reduce this artifact, we apply a local tapered trend correction to the wrapped segment after shifting. Fix a positive integer $\ell \in [2, 2\tau]$, which determines the number of observations used to estimate the local trend. For a given h , write $\mathbf{z} = S_h \mathbf{v}$. When $h \in [\ell, \tau - \ell]$, compute the least-squares coefficients for the terminal part of the wrapped

segment by

$$\arg \min_{\alpha, \beta} \sum_{t=h-\ell+1}^h \left(z_t - \alpha - \beta \cdot \frac{t-h}{\ell} \right)^2 = \arg \min_{\alpha, \beta} \sum_{t=\tau-\ell+1}^{\tau} \left(v_t - \alpha - \beta \cdot \frac{t-\tau}{\ell} \right)^2 := (\hat{\alpha}_W, \hat{\beta}_W),$$

and the least-squares coefficients for the initial part of the leading segment by

$$\arg \min_{\alpha, \beta} \sum_{t=h+1}^{h+\ell} \left(z_t - \alpha - \beta \cdot \frac{t-h}{\ell} \right)^2 = \arg \min_{\alpha, \beta} \sum_{t=1}^{\ell} \left(v_t - \alpha - \beta \cdot \frac{t}{\ell} \right)^2 := (\hat{\alpha}_L, \hat{\beta}_L).$$

For $h \in [\ell, \tau - \ell]$, we compute the trend-corrected shifted waveform that connects the leading segment in the shifted segment with a tapered version of the wrapped segment. The correction is therefore applied only near the splice due to shifting and is tapered continuously to zero away from the splice. For any fixed $c_0 \geq 1$, the tapering is applied on a window of size $c_0 \ell$ based on linear weights $w(a) = 1 - a/(\ell_h - 1)$ for $a = 0, \dots, \ell_h - 1$ where $\ell_h = \min(c_0 \ell, h)$. We compute the trend-corrected shifted waveform as

$$(C_h \mathbf{v})_t = \begin{cases} z_t + w(h-t) \{ (\hat{\alpha}_L - \hat{\alpha}_W) - (\hat{\beta}_L - \hat{\beta}_W)(h-t)\ell^{-1} \}, & h - \ell_h + 1 \leq t \leq h, \\ z_t, & \text{otherwise.} \end{cases}$$

When $h < \ell$ or $h > \tau - \ell$, we set $C_h \mathbf{v} = S_h \mathbf{v}$. The map C_h is linear and is of the form $C_h = (I_\tau + R_h)S_h$, where R_h has rank at most two. Though theoretically, $\ell \geq 2$ works. In practice, we choose ℓ to be at least 20.

For a matrix $\mathbf{V} = (\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(p)}) \in \mathbb{R}^{\tau \times p}$, we apply the same shift and the trend correction separately to each coordinate, so that $C_h \mathbf{V} = (C_h \mathbf{v}^{(1)}, \dots, C_h \mathbf{v}^{(p)})$. The local trend coefficients may differ across coordinates, but the shift h is common to all coordinates.

For two centered waveforms $\mathbf{u}, \mathbf{v} \in \mathbb{R}^{\tau \times p}$, define the directed shift-adjusted dissimilarity

at shift h by

$$d_h(\mathbf{u}, \mathbf{v}) = \frac{1}{\tau} \sum_{r=1}^p \|\mathbf{u}^{(r)} - C_h \mathbf{v}^{(r)}\|_2^2 = \frac{1}{\tau} \|\mathbf{u} - C_h \mathbf{v}\|_F^2.$$

Let $\mathcal{H}_b = \{0, b, 2b, \dots, \lfloor (\tau - 1)/b \rfloor b\}$ denote the grid of candidate shifts, where $b \geq 1$ is a user-specified step size. The final pairwise dissimilarity between $\mathbf{u}$ and $\mathbf{v}$ is $d(\mathbf{u}, \mathbf{v}) = \min_{h \in \mathcal{H}_b} \{d_h(\mathbf{u}, \mathbf{v}), d_h(\mathbf{v}, \mathbf{u})\}$. Thus, for each orientation, the minimization over h is performed after the squared errors have been added across all coordinates. Consequently, the selected shift is common across coordinates and minimizes their cumulative squared error. This symmetrization allows either waveform to be shifted and trend-corrected relative to the other.

For each pair (i, j) , we compute $d_{ij} = d(\tilde{\mathbf{y}}_i, \tilde{\mathbf{y}}_j)$, and form the pairwise dissimilarity matrix $D = (d_{ij})_{1 \leq i, j \leq n}$. Given a prespecified number of clusters K , we apply single-linkage hierarchical clustering to D and cut the resulting dendrogram to obtain K clusters. Let $\hat{I} = (\hat{I}_1, \dots, \hat{I}_n)$ denote the resulting cluster assignment vector, where $\hat{I}_i \in \{1, \dots, K\}$ is the estimated cluster label of waveform i .

When the number of clusters is unknown, we estimate it using the prediction strength methodology in Tibshirani and Walther (2005). For each candidate K , the sample is repeatedly divided into training and validation subsets, both subsets are clustered into K groups, and the stability of validation-set co-memberships under assignment to the training clusters is measured. We select the largest K whose average prediction strength exceeds a prespecified threshold (0.80 in our simulations), subject to a minimum cluster-size constraint.

2.1 Implementation and Software

Complexity and timing comparison;

Algorithm 1 TC-CyclicClust($\mathbb{X}, \omega, K, b, \ell$): Trend-Corrected Cyclic Clustering

Require: Waveforms $\mathbb{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subset \mathbb{R}^{T \times p}$; rolling-average window length ω ; number of clusters K ; shift-grid step size b ; local trend window length ℓ .

Ensure: Estimated cluster labels $\hat{I} = (\hat{I}_1, \dots, \hat{I}_n)$.

- 1: Set $\tau = T - \omega + 1$.
 - 2: **for** $i = 1, \dots, n$ **do**
 - 3: Compute the coordinatewise rolling-average of the waveforms: $\mathbf{y}_i = A_\omega \mathbf{x}_i \in \mathbb{R}^{\tau \times p}$.
 - 4: Center each coordinate of the waveform: $\tilde{\mathbf{y}}_i = \mathbf{y}_i - \mathbf{1}_\tau \bar{\mathbf{y}}_i^\top$, $\bar{\mathbf{y}}_i = \tau^{-1} \mathbf{y}_i^\top \mathbf{1}_\tau$.
 - 5: **end for**
 - 6: Define the candidate shift grid: $\mathcal{H}_b = \{0, b, 2b, \dots, \lfloor (\tau - 1)/b \rfloor b\}$.
 - 7: **for** $1 \leq i < j \leq n$ **do**
 - 8: **for** each $h \in \mathcal{H}_b$ **do**
 - 9: Compute the trend-corrected shifted waveforms $C_h \tilde{\mathbf{y}}_i$ and $C_h \tilde{\mathbf{y}}_j$.
 - 10: Compute the directed dissimilarities
 - 11: **end for**
 - 12: Set: $d_{ij} = d_{ji} = \min_{h \in \mathcal{H}_b} \{d_h(\tilde{\mathbf{y}}_i, \tilde{\mathbf{y}}_j), d_h(\tilde{\mathbf{y}}_j, \tilde{\mathbf{y}}_i)\}$.
 - 13: **end for**
 - 14: Set $d_{ii} = 0$ for $i = 1, \dots, n$, and form $D = (d_{ij})_{1 \leq i, j \leq n}$.
 - 15: Apply single-linkage hierarchical clustering to the dissimilarity matrix D .
 - 16: Cut the resulting dendrogram to obtain K clusters.
 - 17: **return** $\hat{I} = (\hat{I}_1, \dots, \hat{I}_n)$.
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3 Theory

To understand the theoretical properties of the proposed algorithm, we consider the additive signal-plus-noise model

$$\mathbf{x}_i^{(j)} = \mathbf{g}_i^{(j)} + \sigma_{j,i} \boldsymbol{\epsilon}_i^{(j)}, \quad j = 1, \dots, p \text{ and } i = 1, \dots, n,$$

where $\mathbf{g}_i = (\mathbf{g}_i^{(1)}, \dots, \mathbf{g}_i^{(p)}) \in \mathbb{R}^{T \times p}$ is the deterministic signal component, the noise vectors $\boldsymbol{\varepsilon}_i^{(r)} \sim N(\mathbf{0}, I_T)$, $r = 1, \dots, p$, and are independent across both i and r . Let $I_i \in \{1, \dots, K\}$ denote the true class label of observation i , and define the true clusters $\mathcal{C}_a = \{i : I_i = a\}$, $a = 1, \dots, K$. Throughout this section, we assume that K is known and each true cluster contains at least two observations. To facilitate rigorous theoretical proofs, we assume that the noise variances are bounded:

$$0 < \underline{\sigma}^2 \leq \inf_{1 \leq i \leq n, 1 \leq r \leq p} \sigma_{r,i}^2 \leq \sup_{1 \leq i \leq n, 1 \leq r \leq p} \sigma_{r,i}^2 \leq \bar{\sigma}^2 < \infty.$$

In the homoskedastic case, $\sigma_{r,i} = \sigma$ for every $i = 1, \dots, n$ and $r = 1, \dots, p$.

Henceforth, we consider ℓ, c_0 fixed. Then for any ω -length rolling average window, the centered smoothed waveform is:

$$\tilde{\mathbf{y}}_i = \tilde{\mathbf{f}}_i + \tilde{\boldsymbol{\xi}}_i \text{ where } \tilde{\mathbf{f}}_i = P_\tau A_\omega \mathbf{g}_i \text{ and } \tilde{\boldsymbol{\xi}}_i = (\sigma_{1,i} P_\tau A_\omega \boldsymbol{\varepsilon}_i^{(1)}, \dots, \sigma_{p,i} P_\tau A_\omega \boldsymbol{\varepsilon}_i^{(p)}).$$

Since all these smoothed centered waveforms have length τ in each coordinate, all circular shifts below act on $\mathbb{R}^\tau$ coordinatewise, with the same shift applied to all coordinates.

For an ordered pair (i, j) and $h \in \mathcal{H}_b$, let $\boldsymbol{\Delta}_{ijh} = \tilde{\mathbf{f}}_i - C_h \tilde{\mathbf{f}}_j$, and $\boldsymbol{\eta}_{ijh} = \tilde{\boldsymbol{\xi}}_i - C_h \tilde{\boldsymbol{\xi}}_j$, and define their squared Frobenius norms and cross-product by

$$A_\omega(i, j, h) = \|\boldsymbol{\Delta}_{ijh}\|_F^2, \quad B_\omega(i, j, h) = \|\boldsymbol{\eta}_{ijh}\|_F^2, \quad C_\omega(i, j, h) = 2\langle \boldsymbol{\Delta}_{ijh}, \boldsymbol{\eta}_{ijh} \rangle_F.$$

Then, the directed empirical distance between the waveforms i and j decomposes as

$$d_{\omega,h}(i, j) = A_\omega(i, j, h) + B_\omega(i, j, h) + C_\omega(i, j, h), \quad (1)$$

Note that, for this section we consider the non-normalized distances instead of the normalized average distance. The symmetric minimal distance over the grid is:

$$\rho_\omega(i, j) = \min_{(u, v) \in \{(i, j), (j, i)\}} \min_{h \in \mathcal{H}_b} d_{\omega, h}(u, v). \quad (2)$$

Here the minimization over h is outside the sum over coordinates contained in the Frobenius norm. Thus, a single shift minimizes the cumulative squared error across all coordinates. For any unordered pair $\{i, j\}$, the signal-only distance is

$$\eta_\omega(i, j) = \min_{(u, v) \in \{(i, j), (j, i)\}} \min_{h \in \mathcal{H}_b} A_\omega(u, v, h). \quad (3)$$

Then, the between-cluster signal separation is given by

$$\mathcal{B}_\omega = \min_{1 \leq a < b \leq K} \min_{i \in \mathcal{C}_a, j \in \mathcal{C}_b} \eta_\omega(i, j). \quad (4)$$

For a finite vertex set V , let $\mathbb{T}(V)$ be the set of spanning trees on V . Consider the following within-cluster distance:

$$\mathcal{W}_\omega = \max_{1 \leq a \leq K} \min_{\mathcal{T} \in \mathbb{T}(\mathcal{C}_a)} \max_{\{i, j\} \in E(\mathcal{T})} \eta_\omega(i, j). \quad (5)$$

For each true cluster, this is the largest edge weight in a minimum spanning tree formed using the signal distances $\eta_\omega(i, j)$, followed by a maximum over the K true clusters. Note that, $\mathcal{W}_\omega \leq \mathcal{W}_\omega^*$ where $\mathcal{W}_\omega^* = \max_a \max\{\eta_\omega(i, j) : i, j \in \mathcal{C}_a, i \neq j\}$ is the maximal within-cluster diameter. If $\mathcal{B}_\omega$ is much larger than $\mathcal{W}_\omega$, then the signal gap will help us to correctly identify the clusters. We next rigorously state this result. For this purpose we introduce some notations first. Let $\kappa = 8c_0^{1/2}(1 + c_0^2)^{1/2}$ and $\lambda = 1 + (1 + \kappa)^2$. As

shown in Lemma 5, the coordinatewise correction operators used in **TC-CyclicClust** satisfy: $\text{rank}(R_h) \leq 2$, $\sup_{h \in \mathcal{H}_b} \|R_h\|_F \leq \kappa$ and $\sup_{h \in \mathcal{H}_b} \|C_h\|_{\text{op}} \leq 1 + \kappa$. The same operator is applied separately to every coordinate. Equivalently, after vectorization the aggregate operator is $I_p \otimes C_h$, whose operator norm remains $\|C_h\|_{\text{op}}$. Let $\Sigma_\omega = P_\tau A_\omega A_\omega^\top P_\tau$ and $s_\omega = \text{tr}(\Sigma_\omega)$. Then, $(\tau\omega^{-1} - 1)_+ \leq s_\omega \leq \tau\omega^{-1}$, and $\|\Sigma_\omega\|_{\text{op}} \leq 1$. Define the following two quantities that will be used to characterize the signal gap needed by our algorithm for perfect classification:

$$\underline{\mu}_\omega = s_\omega + \left(s_\omega - 2\sqrt{2}\kappa\right)_+ \text{ and } \bar{\mu}_\omega = 2s_\omega + 2\sqrt{2}\kappa + \kappa^2. \quad (6)$$

Let $N_\omega = n(n-1)|\mathcal{H}_b| \leq n(n-1)(T-\omega+1)$ and define $c_\omega(q) = \log(N_\omega/q)$ for any $q \in (0, 1)$. We probabilistically bound the signal-adaptive cross-term coefficients in (1) uniformly by the following multiplier of the Frobenius norm of their respective signal gap:

$$r_\omega(q) = 2\sqrt{2}\bar{\sigma}\lambda^{1/2}(c_\omega(q) + \log 2)^{1/2}. \quad (7)$$

We use probabilistic bounds on the the pure-noise terms $B_\omega(i, j, h)$ in the decomposition in (1). They are uniformly upper and lower bounded respectively by the following two terms:

$$U_{B,\omega}(q) = \bar{\sigma}^2 \left\{ p\bar{\mu}_\omega + 2\lambda c_\omega(q) + 2\sqrt{p\lambda\bar{\mu}_\omega c_\omega(q)} \right\}, \quad L_{B,\omega}(q) = \left[\sigma^2 p\underline{\mu}_\omega - 2\bar{\sigma}^2 \sqrt{p\lambda\bar{\mu}_\omega c_\omega(q)} \right]_+. \quad (8)$$

We next state the exact recovery result for our single linkage based hierarchical clustering algorithm **TC-CyclicClust**.

Theorem 1. *If the between-cluster and within-cluster signal gap satisfy the following for*

some $\delta \in (0, 1)$:

$$\mathcal{B}_\omega - \mathcal{W}_\omega - r_\omega(\delta/3)\{\sqrt{\mathcal{B}_\omega} + \sqrt{\mathcal{W}_\omega}\} \geq U_{B,\omega}(\delta/3) - L_{B,\omega}(\delta/3), \quad (9)$$

then, the clustering labels $\hat{I}_1, \dots, \hat{I}_n$ produced by **CyclicClust** with window length ω and any $b \geq 1$ satisfy

$$\mathbb{P}\left(\min_{\pi \in \Pi_K} \sum_{i=1}^n \mathbf{1}\{\hat{I}_i \neq \pi(I_i)\} = 0\right) \geq 1 - \delta,$$

where Π_K denotes the set of permutations of $\{1, \dots, K\}$.

In the homoskedastic case, when $p\bar{\mu}_\omega > 2\sqrt{p\lambda\bar{\mu}_\omega c_\omega(\delta/3)}$, then the right side in (9) can be further simplified as

$$U_{B,\omega}(\delta/3) - L_{B,\omega}(\delta/3) = \sigma^2 \left[p(\bar{\mu}_\omega - \underline{\mu}_\omega) + 4\sqrt{p\lambda\bar{\mu}_\omega c_\omega(\delta/3)} + 2\lambda c_\omega(\delta/3) \right]. \quad (10)$$

In our application T is large. Thus, we consider an asymptotic regime where $T \rightarrow \infty$, $\log n/T \rightarrow 0$, and p is fixed. For $\delta_T = T^{-2}$, let $\gamma = c_\omega(\delta_T/3) + \log 2$. Since $N_\omega \leq n(n-1)(T-\omega+1)$, we have $\gamma \asymp \log n + \log T$. We further assume that the window size satisfies $\omega\gamma T^{-1} \rightarrow 0$ as $T \rightarrow \infty$. Using the above result, it follows that, in this asymptotic regime, the probability of any misclassification by **TC-CyclicClust** is at most T^{-2} when the signal gap satisfies

$$\mathcal{B}_\omega - \mathcal{W}_\omega - 2\sqrt{2}\lambda^{1/2}\gamma^{1/2}\sigma\left(\sqrt{\mathcal{B}_\omega} + \sqrt{\mathcal{W}_\omega}\right) \geq 4\sqrt{6}\lambda^{1/2}\sigma^2\sqrt{\gamma p\omega^{-1}T}. \quad (11)$$

We provide the detailed proof of Theorem 1 and (3) in Section 6. Note that λ is a constant. In particular, when $c_0 = 1$, $\lambda = 130 + 16\sqrt{2}$. Ignoring constants, the leading pure-noise separation scale required by our cyclic-shift-based, trend-corrected clustering procedure is

$$\sigma^2 \{p (\log n + \log T) T \omega^{-1}\}^{1/2}.$$

All theoretical distances above are sums of squared differences over both time points and coordinates. Replacing them by mean squared differences over time divides all squared-distance quantities by the common factor τ . Thus, normalization leaves the pairwise ordering, the single-linkage dendrogram, and the recovery condition unchanged after the corresponding rescaling.

Note that, in the right side of the dimension p is due to the bounds on the pure-noise terms in the decomposition (1). Bounding the p -dimensional aggregated error which is related to the trace of its covariance which is proportional to p , while its squared Frobenius norm is also proportional to p , yielding stochastic fluctuation terms of order $p^{1/2}$. In contrast, the logarithmic factor arises from a union bound over observation pairs and candidate shifts, whose number does not increase with p because the same shift is used across all coordinates. We next show that the two principal stochastic terms in the sufficient recovery condition cannot, in general, be removed. The result is a worst-case, algorithm-specific lower bound.

Lemma 1. *Consider the homoskedastic case in which $\sigma_{r,i} = \sigma$ for every $i = 1, \dots, n$ and $r = 1, \dots, p$, and suppose that $\omega/T \rightarrow 0$ as $T \rightarrow \infty$. Consider running **TC-CyclicClust** with K equal to the true number of clusters and with fixed b , ℓ , and c_0 . Then there exist universal constants $c_1, c_2 > 0$ such that, for all sufficiently large T and every $0 \leq \mathcal{W} \leq \mathcal{B} < \infty$, if*

$$\mathcal{B} - \mathcal{W} \leq c_1 \left\{ \sigma \sqrt{\mathcal{B} + \mathcal{W}} + \sigma^2 \sqrt{\frac{p\tau}{\omega}} \right\},$$

*there exists a valid homoskedastic p -dimensional signal configuration satisfying $\mathcal{B}_\omega = \mathcal{B}$ and $\mathcal{W}_\omega = \mathcal{W}$ for which the probability of at least one misclassification by **TC-CyclicClust** is at least c_2 .*

The result establishes worst-case sharpness of the two basic fluctuation scales $\sigma\sqrt{\mathcal{B}_\omega + \mathcal{W}_\omega}$ and $\sigma^2\sqrt{pT/\omega}$. The first term is generated by the signal–noise cross products. The second is generated by fluctuations of the quadratic pure-noise terms, which add over the p independent coordinates and therefore have standard deviation of order $\sqrt{p}$. Since $\tau/T \rightarrow 1$ whenever $\omega = o(T)$, there are valid homoskedastic configurations for which

$$\mathcal{B}_\omega - \mathcal{W}_\omega \lesssim \sigma\sqrt{\mathcal{B}_\omega + \mathcal{W}_\omega} + \sigma^2\sqrt{\frac{pT}{\omega}}$$

leads to a misclassification probability bounded away from zero. The result does not establish necessity of the $\sqrt{\log(nT)}$ factor in the uniform sufficient condition.

Next, note that the factor $\omega^{-1/2}$ in the stochastic term of (3) should not be interpreted as saying that a larger rolling-average window is always preferable. Both $\mathcal{B}_\omega$ and $\mathcal{W}_\omega$ depend on ω , and smoothing can attenuate the signal gap. To understand this tradeoff consider a simple set-up where we have localized rectangular contrast. Let $I = \{s+1, \dots, s+L\} \subset \{1, \dots, T\}$, and suppose that two signals satisfy $\mathbf{g}_1 - \mathbf{g}_2 = \mathbf{1}_I \mathbf{a}^\top$, where $\mathbf{1}_I \in \mathbb{R}^T$ is the indicator vector of I . For a given ω , assume that the contrast is separated from the two endpoints so that $s \geq \omega - 1, s+L \leq T - \omega + 1$. Define the uncentered and centered squared signal contrasts by $G_\omega^{\text{unc}} = \|A_\omega(\mathbf{g}_1 - \mathbf{g}_2)\|_F^2$, $G_\omega^{\text{cen}} = \|P_\tau A_\omega(\mathbf{g}_1 - \mathbf{g}_2)\|_F^2$, where $\tau = T - \omega + 1$. The following result shows the rolling average attenuation on such a rectangular contrast.

Lemma 2. *The centered squared signal contrasts is given by:*

$$G_\omega^{\text{cen}} = \|\mathbf{a}\|_2^2 \left\{ q_L(\omega) - \frac{L^2}{T - \omega - 1} \right\} \text{ where, } q_L(\omega) = \begin{cases} L - \frac{\omega}{3} + \frac{1}{3\omega}, & 1 \leq \omega \leq L, \\ \frac{L^2}{\omega} - \frac{L^3 - L}{3\omega^2}, & \omega \geq L, \end{cases} \quad (12)$$

Note that, $q_L(1) = L$, and G_ω^{cen} decrease as ω increases and the shape function satisfies

$$\arg \max_{\omega \in \{1,2,\dots\}} \sqrt{\omega} q_L(\omega) = L.$$

If the zero shift is signal-optimal in (3), then $\eta_\omega(1,2) = G_\omega^{\text{cen}}$. Consequently, in the two-cluster configuration in which every observation in the first cluster has signal $\mathbf{g}_1$ and every observation in the second cluster has signal $\mathbf{g}_2$, one has $\mathcal{W}_\omega = 0$ and $\mathcal{B}_\omega = G_\omega^{\text{cen}}$.

Next, note that the factor $\omega^{-1/2}$ in the stochastic term of (3) should not be interpreted as implying that a larger rolling-average window is always preferable. Both $\mathcal{B}_\omega$ and $\mathcal{W}_\omega$ depend on ω , and smoothing can attenuate the signal gap. Indeed, the signal-dependent part of the left side of (9) can be written as

$$\mathcal{B}_\omega - \mathcal{W}_\omega - r_\omega(\delta/3) \left(\sqrt{\mathcal{B}_\omega} + \sqrt{\mathcal{W}_\omega} \right) = \left(\sqrt{\mathcal{B}_\omega} + \sqrt{\mathcal{W}_\omega} \right) \left\{ \sqrt{\mathcal{B}_\omega} - \sqrt{\mathcal{W}_\omega} - r_\omega(\delta/3) \right\}.$$

Thus, the relevant deterministic separation on the distance scale is $\sqrt{\mathcal{B}_\omega} - \sqrt{\mathcal{W}_\omega}$, which may decrease when ω is too large.

To illustrate this tradeoff, consider a localized rectangular contrast. Let $I = \{s + 1, \dots, s + L\} \subset \{1, \dots, T\}$, and suppose that two signals satisfy

$$\mathbf{g}_1 - \mathbf{g}_2 = \mathbf{1}_I \mathbf{a}^\top,$$

where $\mathbf{1}_I \in \mathbb{R}^T$ is the indicator vector of I and $\mathbf{a} \in \mathbb{R}^p$ is the height contrast. For a given ω , assume that the rectangular contrast is sufficiently far from the endpoints so that

$$s \geq \omega - 1, \quad s + L \leq T - \omega + 1.$$

Define the uncentered and centered squared signal contrasts by

$$G_\omega^{\text{unc}} = \|A_\omega(\mathbf{g}_1 - \mathbf{g}_2)\|_F^2, \quad G_\omega^{\text{cen}} = \|P_\tau A_\omega(\mathbf{g}_1 - \mathbf{g}_2)\|_F^2,$$

where $\tau = T - \omega + 1$. The following result quantifies the attenuation caused by rolling averaging.

Lemma 3. *For the rectangular contrast above,*

$$G_{\omega}^{\text{unc}} = \|\mathbf{a}\|_2^2 q_L(\omega), \quad G_{\omega}^{\text{cen}} = \|\mathbf{a}\|_2^2 \tilde{q}_L(\omega), \quad (13)$$

where

$$\tilde{q}_L(\omega) = q_L(\omega) - \frac{L^2}{\tau}$$

and

$$q_L(\omega) = \begin{cases} L - \frac{\omega}{3} + \frac{1}{3\omega}, & 1 \leq \omega \leq L, \\ \frac{L^2}{\omega} - \frac{L^3 - L}{3\omega^2}, & \omega \geq L. \end{cases} \quad (14)$$

Moreover, $q_L(\omega)$ and $\tilde{q}_L(\omega)$ are non-increasing over the feasible window lengths, $q_L(1) = L$,

and

$$\arg \max_{\omega \in \{1, 2, \dots\}} \sqrt{\omega} q_L(\omega) = L.$$

The preceding calculation extends directly to the within- and between-cluster signal quantities used in our recovery result. Suppose that, after signal-optimal alignment, the signals have the form

$$\mathbf{g}_i = \mathbf{g}_0 + \mathbf{1}_I \boldsymbol{\theta}_i^\top, \quad i = 1, \dots, n,$$

where $\mathbf{g}_0 \in \mathbb{R}^{T \times p}$ is common to all observations and $\boldsymbol{\theta}_i \in \mathbb{R}^p$ is an observation-specific amplitude vector. Assume that the zero shift attains the minimum in (3) for every pair. Define the between-cluster amplitude separation by

$$\Delta_B = \min_{1 \leq k < l \leq K} \min_{i \in \mathcal{C}_k, j \in \mathcal{C}_l} \|\boldsymbol{\theta}_i - \boldsymbol{\theta}_j\|_2$$

and the within-cluster amplitude radius by

$$\Delta_W = \max_{1 \leq k \leq K} \min_{\mathcal{T} \in \mathbb{T}(\mathcal{C}_k)} \max_{\{i,j\} \in E(\mathcal{T})} \|\boldsymbol{\theta}_i - \boldsymbol{\theta}_j\|_2.$$

Lemma 4. *Under the preceding rectangular signal model,*

$$\eta_\omega(i, j) = \tilde{q}_L(\omega) \|\boldsymbol{\theta}_i - \boldsymbol{\theta}_j\|_2^2$$

for every $i \neq j$. Consequently,

$$\mathcal{B}_\omega = \tilde{q}_L(\omega) \Delta_B^2, \quad \mathcal{W}_\omega = \tilde{q}_L(\omega) \Delta_W^2. \quad (15)$$

In particular,

$$\mathcal{B}_\omega - \mathcal{W}_\omega = \tilde{q}_L(\omega) (\Delta_B^2 - \Delta_W^2) \quad (16)$$

and

$$\sqrt{\mathcal{B}_\omega} - \sqrt{\mathcal{W}_\omega} = \sqrt{\tilde{q}_L(\omega)} (\Delta_B - \Delta_W). \quad (17)$$

Therefore, under this model, the separation condition in Theorem 1 becomes

$$\tilde{q}_L(\omega) (\Delta_B^2 - \Delta_W^2) - r_\omega(\delta/3) \sqrt{\tilde{q}_L(\omega)} (\Delta_B + \Delta_W) \geq U_{B,\omega}(\delta/3) - L_{B,\omega}(\delta/3).$$

Equivalently, whenever $\tilde{q}_L(\omega) > 0$ and $\Delta_B + \Delta_W > 0$, this condition can be written as

$$\sqrt{\tilde{q}_L(\omega)} (\Delta_B - \Delta_W) \geq r_\omega(\delta/3) + \frac{U_{B,\omega}(\delta/3) - L_{B,\omega}(\delta/3)}{\sqrt{\tilde{q}_L(\omega)} (\Delta_B + \Delta_W)}.$$

When $L = o(T)$ and $\omega = O(L)$, the centering correction L^2/τ is of smaller order

than $q_L(\omega)$. If the signal–noise cross-product term is not dominant, the leading balance between the signal gap and the pure-noise fluctuation is therefore governed by $\sqrt{\omega} q_L(\omega)$. By Lemma 3, this quantity is maximized at $\omega = L$. In contrast, when $\omega \gg L$, $q_L(\omega) \asymp L^2/\omega$, so the signal gap decreases at rate ω^{-1} , whereas the leading stochastic term decreases only at rate $\omega^{-1/2}$. Thus, a useful rolling-average window should generally be of the same order as the width L of the distinguishing rectangular feature and should not be taken arbitrarily large.

As a special case, suppose that all observations within each cluster have the same signal. For two clusters whose amplitude vectors differ by $\mathbf{a}$, one has $\Delta_W = 0$, $\Delta_B = \|\mathbf{a}\|_2$, and hence

$$\mathcal{W}_\omega = 0, \quad \mathcal{B}_\omega = G_\omega^{\text{cen}} = \|\mathbf{a}\|_2^2 \tilde{q}_L(\omega).$$

Proof of Lemma 14. To verify (14), let c_t be the number of indices in I that fall in the t th rolling window. Then the t th row of $A_\omega(\mathbf{g}_1 - \mathbf{g}_2)$ is $(c_t/\omega)\mathbf{a}^\top$. If $\omega \leq L$, the nonzero overlap counts are

$$1, 2, \dots, \omega - 1, \underbrace{\omega, \dots, \omega}_{L-\omega+1 \text{ times}}, \omega - 1, \dots, 2, 1,$$

whereas, if $\omega \geq L$, they are

$$1, 2, \dots, L - 1, \underbrace{L, \dots, L}_{\omega-L+1 \text{ times}}, L - 1, \dots, 2, 1.$$

Thus, using $\sum_{k=1}^m k^2 = m(m+1)(2m+1)/6$,

$$\omega^2 q_L(\omega) = \begin{cases} (L - \omega + 1)\omega^2 + 2 \sum_{k=1}^{\omega-1} k^2, & \omega \leq L, \\ (\omega - L + 1)L^2 + 2 \sum_{k=1}^{L-1} k^2, & \omega \geq L, \end{cases}$$

which gives (14). Also, $\mathbf{1}_\tau^\top A_\omega(\mathbf{g}_1 - \mathbf{g}_2) = L\mathbf{a}^\top$, and the projection identity $\|P_\tau M\|_F^2 = \|M\|_F^2 - \tau^{-1}\|\mathbf{1}_\tau^\top M\|_2^2$ gives (??). For its continuous extension, the derivative of q_L is $-1/3 - 1/(3\omega^2) < 0$ when $\omega \leq L$, and is

$$-\frac{L^2}{\omega^2} + \frac{2(L^3 - L)}{3\omega^3} < 0$$

when $\omega \geq L$. This proves the stated monotonicity. To prove (??), set $f_L(x) = \sqrt{x} q_L(x)$.

For $1 \leq x \leq L$,

$$f'_L(x) = \frac{3Lx - 3x^2 - 1}{6x^{3/2}},$$

which is positive on $[1, L - 1]$ when $L \geq 2$. Moreover,

$$f_L(L)^2 - f_L(L - 1)^2 = \frac{3L^3 - 4L^2 + L - 1}{9L(L - 1)} > 0.$$

For $x \geq L$,

$$f'_L(x) = \frac{L^3 - L - L^2x}{2x^{5/2}} < 0.$$

Thus the maximum over the positive integer window lengths is attained at $\omega = L$; the case $L = 1$ is immediate.

For comparison, let A_ω^{block} denote nonoverlapping block averaging. If ω divides L and the interval I is a union of complete blocks, then

$$\|A_\omega^{\text{block}}(\mathbf{g}_1 - \mathbf{g}_2)\|_F^2 = \frac{L}{\omega} \|\mathbf{a}\|_2^2. \quad (18)$$

The difference between (18) and (14) is due to the overlapping outputs of the rolling-average operator.

The exact calculation gives the simpler approximations

$$G_{\omega}^{\text{unc}} = \begin{cases} \|\mathbf{a}\|_2^2 \left\{ L - \frac{\omega}{3} + O(\omega^{-1}) \right\}, & \omega \leq L, \\ \|\mathbf{a}\|_2^2 \frac{L^2}{\omega} \left\{ 1 + O\left(\frac{L}{\omega}\right) \right\}, & \omega/L \rightarrow \infty. \end{cases} \quad (19)$$

If $L = o(T)$ and $\omega = o(T)$, the centering correction in (??) is of smaller order in the corresponding localized-pulse regimes. Thus, in the univariate case, a pulse-width L and height difference a give the unsmoothed squared contrast La^2 , the rolling-average contrast is approximately $(L - \omega/3)a^2$ for $\omega \leq L$, and it is approximately L^2a^2/ω when ω is much larger than L .

This attenuation explains why the $\omega^{-1/2}$ stochastic scale in (3) does not favor arbitrarily large windows. In the idealized rectangular setting with $\mathcal{W}_{\omega} = 0$, and ignoring the weak dependence of the logarithmic factor γ on ω , the signal relative to the leading pure-noise fluctuation is proportional to

$$\frac{G_{\omega}^{\text{unc}}}{\sqrt{T/\omega}} = \frac{\|\mathbf{a}\|_2^2}{\sqrt{T}} \sqrt{\omega} q_L(\omega). \quad (20)$$

The same balance is obtained for the normalized distances because both the signal contrast and the stochastic fluctuation are divided by the common factor τ . By (??), this proxy is maximized at $\omega = L$. For $\omega \gg L$, it decreases as $L^2/\sqrt{\omega}$: the signal contrast then decays as ω^{-1} , which is faster than the $\omega^{-1/2}$ reduction in the stochastic fluctuation. The signal–noise cross-product term can favor a somewhat smaller window because it also becomes harder to dominate as the signal contrast is attenuated. Hence, the appropriate rolling window is generally of the same order as, and need not exceed, the width of the localized feature that separates the clusters. In particular, choosing ω much larger than that feature width

constitutes over-smoothing and can make the recovery condition harder rather than easier to satisfy.

4 Simulation Studies

In this section, we first compare the performance of **TC-CyclicClust** with that of other established curve clustering methods. We show that our proposed method performs favorably relative to these competitors in settings where accounting for cyclic variation is essential. Next, we evaluate the performance of the proposed method across a wide range of signal-to-noise regimes and smoothing levels. Throughout this section, we applied **TC-CyclicClust** with the tuning parameters set to: $b = 1,000$, $c_0 = 2$, and $\ell = 100$. All code used to generate the simulation results is available on GitHub.

4.1 Study I

We generated $n = 100$ independent univariate waveforms observed at $T = 1,000,000$ equally spaced time points. We considered $K = 4$ clusters with 25 curves in each cluster. The ordering of cluster labels was randomly permuted in every replicate. A waveform from cluster g was generated as

$$x_i(t) = a_i s_g([t - \delta_i]_T) + \varepsilon_i(t), \quad t = 0, \dots, T-1, \quad (21)$$

where $[\cdot]_T$ denotes cyclic reduction, δ_i is uniform on $\{0, \dots, T-1\}$, and $\varepsilon_i(t) \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$. If A denotes the nominal signal amplitude in a regime, the individual amplitude was $|a_i| = AU_i$, with $U_i \sim \text{Unif}(0.85, 1.15)$. Its sign and pulse width depended on the cluster: (1) a positive rectangular pulse of width $W_i \sim \text{Unif}(0.10T, 0.12T)$; (2) a negative

rectangular pulse with the same width distribution; (3) a positive rectangular pulse of width $W_i \sim \text{Unif}(0.38T, 0.42T)$; and (4) a negative rectangular pulse with the same width-pulse distribution. Pulses were wrapped cyclically if they crossed the endpoint. Thus, absolute pulse location contains no cluster information; sign and pulse width (equivalently, duty cycle) carry the cluster information.

For each window length ω , the raw waveform was transformed by the rolling average and then centered by subtracting its sample mean. The four choices were $\omega \in \{1, 10^3, 10^4, 10^5\}$.

We considered four regimes with decreasing nominal amplitude A and increasing noise standard deviation σ : strong $(A, \sigma) = (20, 5)$, moderate $(8, 15)$, weak $(3, 30)$, and very weak $(2, 40)$.

We considered four competing methods. (a) Smoothed Euclidean single linkage (Euclid-SL): This baseline computes ordinary squared Euclidean distances between the smoothed, centered curves, without registration, and applies single linkage. It tests whether smoothing alone can overcome phase variation. (b) Shift-invariant K -means (SI- K -means): This method alternates between assigning each curve to the centroid with minimum squared distance after optimization over the cyclic-shift grid and aligning the assigned curves to recompute each centroid. The implementation used eight random starts and at most 35 iterations. (c) SACK (Liu and Yang, 2009), a parametric curve clustering method: We represented each cluster mean with 13 cubic B-splines and modeled subject-specific time translation and vertical displacement as Gaussian random effects. (d) GS-EM, based on Gaffney and Smyth (2004), is a probabilistic regression-mixture approach that jointly learns cluster membership and continuous alignment using an EM algorithm. We used a cyclic-translation specialization matched to the present data-generating mechanism: cluster templates were represented by 31 cubic B-splines, translation was summed over the 1,000-point cyclic shift grid, the curve-specific amplitude scale was integrated out analytically, and generalized EM

updates used the maximum-posterior translation.

We applied all the methods with the number of clusters set to the true value of 4. In Table 1, we report, for each method, smoothing level ω , and signal-to-noise regime, the misclassification error $\text{Err}(\hat{z}, z) = \min_{\pi \in \mathcal{S}_K} n^{-1} \sum_{i=1}^n \mathbf{1}\{\pi(\hat{z}_i) \neq z_i\}$. We use 25 replicates for each case and report the Monte Carlo standard deviation in parentheses. Euclid-SL performs poorly throughout, demonstrating that smoothing alone cannot overcome the global phase variation induced by random pulse locations. SACK remains between 53.5% and 67.2%, consistent with its local, non-cyclic translation model being mismatched to the global cyclic phase variation considered here. GS-EM is less stable, occasionally achieving low error but exhibiting substantial non-monotonicity and large Monte Carlo variability across settings. In contrast, TC-CyclicClust and SI- K -means demonstrate that cyclic alignment coupled with moderate smoothing performs well. TC-CyclicClust is perfect in the strong and moderate regimes for $\omega \leq 10^4$, while in the weak regime its error decreases from 72.9% at $\omega = 1$ to zero at $\omega = 10^3$ and 10^4 . SI- K -means exhibits a similar pattern. Excessive smoothing, however, deteriorates the performance of TC-CyclicClust, with errors increasing to 6.1%–18.8% at $\omega = 10^5$, as smoothing reduces both noise and the shape contrast that distinguishes the clusters.

While Table 1 shows that both SI- K -means and TC-CyclicClust perform well, we next consider a regime in which single linkage is preferable to K -means. We considered $K = 2$ clusters with 50 curves each. Let $q_1, q_2 \in \mathbb{R}^T$ be centered, RMS-orthonormal rectangular-pulse motifs, and let q_0 be a common asymmetric landmark waveform. An unshifted raw curve was generated as

$$u_i(t) = 60q_0(t) + r_{z_i} \{\cos(\theta_i)q_1(t) + \sin(\theta_i)q_2(t)\} + \varepsilon_i(t), \quad (22)$$

Table 1: Misclassification percentage of different clustering methods: mean (standard deviation), based on 25 replicates per cell.

Regime	(A, σ)	ω	TC-CyclicClust	Euclid-SL	SI- K -means	SACK	GS-EM
Strong	(20, 5)	1	0.0 (0.0)	60.7 (10.3)	1.0 (4.8)	55.3 (3.8)	10.5 (19.2)
		10^3	0.0 (0.0)	57.2 (11.6)	1.3 (6.6)	53.9 (5.7)	36.0 (31.5)
		10^4	0.0 (0.0)	58.2 (11.8)	0.0 (0.0)	54.6 (4.3)	57.0 (27.5)
		10^5	6.6 (8.7)	63.2 (8.3)	0.0 (0.0)	56.0 (2.4)	75.0 (0.0)
Moderate	(8, 15)	1	0.0 (0.0)	72.8 (0.5)	48.3 (5.9)	55.7 (4.4)	11.2 (15.4)
		10^3	0.0 (0.0)	58.7 (10.6)	0.0 (0.0)	55.6 (3.6)	33.1 (26.4)
		10^4	0.0 (0.0)	60.0 (11.7)	0.0 (0.0)	54.8 (5.4)	45.0 (28.0)
		10^5	6.1 (10.4)	62.6 (5.5)	0.0 (0.0)	54.1 (3.7)	75.0 (0.0)
Weak	(3, 30)	1	72.9 (0.6)	72.7 (0.7)	68.2 (2.5)	66.5 (2.5)	64.8 (3.3)
		10^3	0.0 (0.0)	66.6 (7.3)	0.0 (0.0)	53.5 (4.4)	8.7 (18.1)
		10^4	0.0 (0.0)	65.3 (7.7)	0.0 (0.0)	54.2 (3.9)	31.7 (32.5)
		10^5	9.2 (9.9)	62.5 (8.6)	0.0 (0.0)	54.3 (5.5)	75.0 (0.0)
Very weak	(2, 40)	1	72.7 (0.7)	72.6 (0.5)	68.2 (2.0)	67.2 (2.0)	66.4 (2.5)
		10^3	6.0 (10.9)	71.3 (2.2)	4.6 (10.7)	54.6 (3.4)	21.6 (30.5)
		10^4	0.0 (0.0)	69.1 (4.2)	0.0 (0.0)	54.5 (4.4)	15.0 (30.6)
		10^5	18.8 (10.6)	65.5 (6.3)	0.0 (0.2)	54.2 (4.9)	75.0 (0.2)

and the input to clustering was $y_i = R_{\delta_i} P_\tau A_\omega u_i$, where R_{δ_i} denotes a cyclic shift of the retained smoothed curve, $(r_1, r_2) = (1, 4)$, and $\varepsilon_i(t) \sim N(0, 150^2)$. Within each cluster, the 50 angles nearly evenly cover $[0, 2\pi)$, with independent $N(0, 0.018^2)$ angular jitter and $\text{Unif}(0.985, 1.015)$ radial jitter. The strong common landmark q_0 contains no cluster information; it only ensures that both methods use a common phase alignment. The resulting clusters are connected but strongly non-convex, providing a setting naturally favorable to linkage-based clustering relative to centroid-based clustering. In Table 2, we report the misclassification error for $T = 10^6$. The proposed single-linkage method, TC-CyclicClust, has 3.9% error at $\omega = 10^3$ and perfectly recovers the two clusters at $\omega = 10^4$, whereas SI- K -means remains near chance across all smoothing levels. This experiment illustrates the advantage of TC-CyclicClust when the aligned cluster geometry is non-convex and therefore poorly represented by cluster centroids.

Table 2: Misclassification percentage (mean (SD)) for non-convex clusters over 25 replicates.

Method	$\omega = 1$	$\omega = 10^3$	$\omega = 10^4$	$\omega = 10^5$
TC-CyclicClust	49.0 (0.0)	3.9 (13.4)	0.0 (0.0)	22.6 (22.0)
SI- K -means	47.0 (2.6)	45.4 (3.2)	45.2 (4.0)	47.0 (2.0)

4.2 Study II

Balanced Design. We first consider a balanced design with $K = 4$ clusters and 25 observations per cluster. Each observation is a five-dimensional waveform measured at $T = 10^6$ equally spaced time points. Let z_i denote the true cluster of subject i . For coordinate r , the raw waveform is generated as

$$x_{it}^{(r)} = I(r \in \mathcal{S}) a_i I\{(t - s_i - \delta_{ir}) \bmod T \in [0, L_i]\} + \varepsilon_{it}^{(r)}, \quad \varepsilon_{it}^{(r)} \stackrel{\text{ind}}{\sim} N(0, \sigma_r^2), \quad (23)$$

where $\mathcal{S}$ is the set of coordinates containing signal, s_i is a uniformly generated cyclic location, L_i is the pulse width, and δ_{ir} is a coordinate-specific alignment perturbation. The four population signals correspond to a narrow peak, a narrow trough, a wide peak, and a wide trough. The subject-specific pulse height is $a_i = q_{z_i} A U_i$, where $q_{z_i} \in \{-1, 1\}$ distinguishes peaks from troughs and $U_i \sim \text{Unif}(0.85, 1.15)$. The narrow-pulse width is drawn uniformly between $0.10T$ and $0.12T$, while the wide-pulse width is drawn between $0.38T$ and $0.42T$.

We consider three multivariate coordinate designs. (a) *Heterogeneous coordinate noise.* All five coordinates contain the same signal at a common cyclic location, while their noise standard deviations are $(0.75\sigma, 1.00\sigma, 1.25\sigma, 1.50\sigma, 1.75\sigma)$. (b) *Coordinate misalignment.* All five coordinates contain the same signal and have noise standard deviation σ , but independently across subjects and coordinates, $\delta_{ir} \in \{-10000, -2500, 0, 2500, 5000\}$. (c) *Spurious signal coordinates.* Only coordinates 1 and 2 contain the rectangular pulse, while

coordinates 3–5 consist entirely of independent $N(0, \sigma^2)$ noise.

These designs respectively assess robustness to heterogeneous noise, a mild violation of the common-shift assumption, and dilution of the signal by uninformative coordinates.

Within each coordinate design, we examine strong, moderate, and weak signal regimes obtained by fixing $A = 20$ and setting $\sigma = 120, 400$, and 650 , respectively. Thus, the rectangular-pulse contrast decreases relative to the background noise from the strong to the weak regime. Each coordinate is smoothed and centered before distances are computed, using rolling-average window lengths $\omega \in \{1, 10^3, 10^4\}$. The multivariate cyclic distance combines the five coordinate-wise contributions and optimizes over a single common cyclic shift. The resulting dissimilarities are clustered by single linkage.

The number of clusters is selected by prediction strength (Tibshirani and Walther, 2005). Candidate values are $k \in \{1, \dots, 10\}$. For each candidate, we use 50 seeded random half-splits and select the largest admissible value whose mean prediction strength is at least 0.80. A candidate cluster is required to contain at least five observations in each half-sample. For settings in which $\hat{K} \neq K$, the misclassification error (MCE) is computed by padding the contingency table with dummy clusters and finding the maximum-overlap one-to-one matching. Consequently, both merging and splitting of clusters are penalized. In addition to MCE and $\Pr(\hat{K} = K)$, we report the adjusted Rand index (ARI) and the MCE conditional on selecting the correct number of clusters (CMCE). ARI is reported on its 0–1 scale, while MCE, CMCE, and the selection probability are reported as percentages. Parenthetical values are Monte Carlo standard deviations based on 25 replicates.

Table 3 shows that the unsmoothed choice $\omega = 1$ is uniformly suboptimal. Its MCE ranges from 68% to 75%, and prediction strength never selects $K = 4$. At $\omega = 10^3$, the strong regime is recovered perfectly in all three coordinate designs. Performance then deteriorates as the signal weakens. In the two-signal-coordinate design, the MCE increases

Table 3: Performance of TY-CYCLICCLUST under balanced multivariate regimes. ARI is reported on its $[0, 1]$ scale; MCE, CMCE, and $\Pr(\hat{K} = K)$ are reported as percentages.

Coordinate design	Regime	ω	ARI	MCE	$\Pr(\hat{K} = K)$	CMCE
Heterogeneous noise	Strong	1	0.000 (0.000)	75.0 (0.0)	0	—
		1000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
		10000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
	Moderate	1	0.000 (0.000)	75.0 (0.0)	0	—
		1000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
		10000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
	Weak	1	0.000 (0.000)	75.0 (0.0)	0	—
		1000	0.000 (0.000)	75.0 (0.0)	0	—
		10000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
Misaligned	Strong	1	0.138 (0.226)	68.0 (11.5)	0	—
		1000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
		10000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
	Moderate	1	0.000 (0.000)	75.0 (0.0)	0	—
		1000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
		10000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
	Weak	1	0.000 (0.000)	75.0 (0.0)	0	—
		1000	0.602 (0.375)	34.0 (29.7)	36	0.0 (0.0)
		10000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
Spurious coordinates	Strong	1	0.000 (0.000)	75.0 (0.0)	0	—
		1000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
		10000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
	Moderate	1	0.000 (0.000)	75.0 (0.0)	0	—
		1000	0.796 (0.348)	17.0 (27.7)	68	0.0 (0.0)
		10000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
	Weak	1	0.000 (0.000)	75.0 (0.0)	0	—
		1000	0.000 (0.000)	75.0 (0.0)	0	—
		10000	0.631 (0.381)	31.0 (30.0)	40	0.0 (0.0)

from zero in the strong regime to 17% in the moderate regime and 75% in the weak regime, while $\Pr(\hat{K} = 4)$ decreases from 100% to 68% and then to zero. In the weak misalignment regime, the corresponding MCE is 34% and the correct number of clusters is selected in 36% of replicates.

The larger window $\omega = 10^4$ gives the best overall performance. It perfectly recovers every balanced configuration except the weak two-signal-coordinate design. In that most difficult balanced setting, the mean ARI is 0.631, the mean MCE is 31%, and $\Pr(\hat{K} = 4) =$

40%. Whenever prediction strength selects the correct number of clusters, CMCE is zero, indicating that the nonzero unconditional errors arise from under-selection of K . We next examine whether this robustness persists when the number of clusters increases and several clusters contain relatively few observations.

Imbalanced Designs. We retain the waveform construction, coordinate designs, pre-processing, and clustering procedure described above, but now consider two imbalanced cluster-size configurations. The first contains $K = 6$ clusters and $n = 90$ curves, with sizes $(20, 20, 20, 10, 10, 10)$. The second contains $K = 10$ clusters and $n = 230$ curves, with five clusters of size 30, three of size 20, and two of size 10.

The clusters are again organized into peak-trough pairs. For $K = 6$, the centers of the three pulse-width pairs are $0.10T$, $0.26T$, and $0.42T$. For $K = 10$, the centers of the five pairs are $0.08T$, $0.165T$, $0.25T$, $0.335T$, and $0.42T$. Conditional on cluster membership, each subject’s pulse width is drawn uniformly within $0.01T$ of the corresponding center. Together with the random pulse height and cyclic location in (23), this ensures that subjects within a cluster are not exact copies.

We fix the pulse amplitude at $A = 20$ and the baseline noise standard deviation at $\sigma = 340$. This corresponds to the intermediate signal-to-noise setting. We use the same parameters as before except prediction strength now searches $k \in \{1, \dots, 14\}$.

As in the balanced regimes, smoothing is essential at this signal-to-noise level. With $\omega = 1$, prediction strength severely under-selects K in every configuration, producing ARI equal to zero. Moderate smoothing with $\omega = 10^3$ is sufficient for perfect recovery in the two six-cluster designs where all coordinates contain signal. In presence of spurious clusters, the six-cluster design still has mean MCE only 3.1%, mean ARI 0.960, and $\Pr(\hat{K} = K) = 96\%$.

The ten-cluster design is more difficult at $\omega = 10^3$. The MCE is 73.9% under heterogeneous coordinate noise, 37.7% under small misalignment, and 80.9% when only two

coordinates contain signal. The corresponding correct-selection probabilities are 0%, 48%, and 0%. The larger window $\omega = 10^4$ gives the best overall performance. It perfectly recovers every configuration except the ten-cluster, two-signal-coordinate design. In that most difficult setting, $\Pr(\hat{K} = K) = 76\%$, the mean ARI is 0.818, and the mean MCE is 17.0%. This deterioration reflects the combined difficulty of several small clusters and dilution of the signal by three pure-noise coordinates.

Whenever prediction strength selects the correct number of clusters, CMCE is zero. Thus, the nonzero unconditional errors arise from under-selection of K , rather than incorrect assignment after the correct value of K has been selected.

Table 4: Performance of TY-CYCLICCLUST under imbalanced multivariate regimes. ARI is reported on its $[0, 1]$ scale; MCE, CMCE, and $\Pr(\hat{K} = K)$ are reported as percentages.

Scenario	Coordinate regime	ω	ARI	MCE	$\Pr(\hat{K} = K)$	CMCE
$K = 6$	Heterogeneous noise	1	0.000 (0.000)	77.8 (0.0)	0	—
		1000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
		10000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
	Misaligned	1	0.000 (0.000)	77.8 (0.0)	0	—
		1000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
		10000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
	Spurious coordinates	1	0.000 (0.000)	77.8 (0.0)	0	—
		1000	0.960 (0.200)	3.1 (15.6)	96	0.0 (0.0)
		10000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
$K = 10$	Heterogeneous noise	1	0.000 (0.000)	87.0 (0.0)	0	—
		1000	0.211 (0.045)	73.9 (3.1)	0	—
		10000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
	Misaligned	1	0.000 (0.000)	87.0 (0.0)	0	—
		1000	0.598 (0.395)	37.7 (37.0)	48	0.0 (0.0)
		10000	1.000 (0.000)	0.0 (0.0)	100	0.0 (0.0)
	Spurious coordinates	1	0.000 (0.000)	87.0 (0.0)	0	—
		1000	0.098 (0.113)	80.9 (7.1)	0	—
		10000	0.818 (0.330)	17.0 (31.0)	76	0.0 (0.0)

5 Discussion and Future Work

Faster algorithm (Fourier based) for online usage; variable selection for high-dimensional signals as this methods suffers if there are too many spurious signal coordinates (cite simulation section example);

6 Proof of Theorem 1

The proof consists of four key steps. First, we establish stability of the cyclic operator C_h and show that it can amplify the noise at most by a constant amount and not by an amount that grows with T .

Lemma 5. *For all $h \in \mathcal{H}_b$ $\text{rank}(R_h) \leq 2$, $\|R_h\|_F \leq \kappa$, and, $\|C_h\|_{\text{op}} \leq 1 + \kappa$, where $\kappa = 8c_0^{1/2}(1 + c_0^2)^{1/2}$.*

For a p -dimensional waveform, the same operator is applied separately to each coordinate. Equivalently, after vectorization, the aggregate correction operator is $I_p \otimes C_h$, and $\|I_p \otimes C_h\|_{\text{op}} = \|C_h\|_{\text{op}} \leq 1 + \kappa$. Thus, the operator-norm bound remains independent of p , although $\text{rank}(I_p \otimes R_h) \leq 2p$ and $\|I_p \otimes R_h\|_F \leq \sqrt{p} \kappa$.

Second, we study the covariance structure of the noise after the impact of the transformations used in Algorithm 1. The transformations are rolling averaging, centering, circular shifting, and local tapered correction. Using the operator bounds from the first step, we obtain uniform upper and lower bounds for the trace of the transformed covariance matrix, as well as uniform bounds for its operator and Frobenius norms.

Let $Q_\omega = A_\omega A_\omega^\top$. Its entries are $(Q_\omega)_{rs} = (\omega - |r - s|)_+ / \omega^2$, so $\text{tr}(Q_\omega) = \tau / \omega$. Jensen's inequality also gives $\|A_\omega\|_{\text{op}} \leq 1$, and hence $\|Q_\omega\|_{\text{op}} \leq 1$. Recall from Section 3 that $\Sigma_\omega = P_\tau Q_\omega P_\tau$ and $s_\omega = \text{tr}(\Sigma_\omega)$. With $\mathbf{u} = \tau^{-1/2} \mathbf{1}_\tau$, one has $P_\tau = I_\tau - \mathbf{u} \mathbf{u}^\top$ and therefore

$s_\omega = \tau/\omega - \mathbf{u}^\top Q_\omega \mathbf{u}$. Since $0 \leq \mathbf{u}^\top Q_\omega \mathbf{u} \leq 1$, this proves $(\tau/\omega - 1)_+ \leq s_\omega \leq \tau/\omega$, while the projection property gives $\|\Sigma_\omega\|_{\text{op}} \leq 1$.

For each i and coordinate $r = 1, \dots, p$, the centered smoothed noise satisfies $\tilde{\xi}_i^{(r)} \sim N(\mathbf{0}, \sigma_{r,i}^2 \Sigma_\omega)$. Hence, for a fixed ordered triple (i, j, h) , $\boldsymbol{\eta}_{ijh}^{(r)} \sim N(\mathbf{0}, \Psi_{\omega,ijh}^{(r)})$, and $\Psi_{\omega,ijh}^{(r)} = \sigma_{r,i}^2 \Sigma_\omega + \sigma_{r,j}^2 C_h \Sigma_\omega C_h^\top$. Because the noises are independent across coordinates, $\text{vec}(\boldsymbol{\eta}_{ijh}) \sim N(\mathbf{0}, \Psi_{\omega,ijh})$, and, $\Psi_{\omega,ijh} = \text{diag}(\Psi_{\omega,ijh}^{(1)}, \dots, \Psi_{\omega,ijh}^{(p)})$. The following result provides bounds on this covariance.

Lemma 6. *Uniformly over all ordered pairs $i \neq j$ and all $h \in \mathcal{H}_b$,*

$$\sigma^2 p \underline{\mu}_\omega \leq \text{tr}(\Psi_{\omega,ijh}) \leq \bar{\sigma}^2 p \bar{\mu}_\omega, \quad \|\Psi_{\omega,ijh}\|_{\text{op}} \leq \bar{\sigma}^2 \lambda, \quad \text{and}, \quad \|\Psi_{\omega,ijh}\|_F^2 \leq \bar{\sigma}^4 p \lambda \bar{\mu}_\omega. \quad (24)$$

Third, these covariance bounds are combined with Gaussian quadratic-form and Gaussian tail inequalities. A union bound over all ordered observation-pair and shift combinations then gives simultaneous high-probability upper and lower bounds for the pure-noise term and a signal-adaptive bound for the signal-noise cross term.

Lemma 7. *For every $q \in (0, 1)$, with $U_{B,\omega}$, $L_{B,\omega}$ and r_ω defined in Section 3, we have*

$$\mathbb{P} \left\{ \sup_{i \neq j} \sup_{h \in \mathcal{H}_b} B_\omega(i, j, h) < U_{B,\omega}(q) \right\} \geq 1 - q, \quad (25)$$

$$\mathbb{P} \left\{ \inf_{i \neq j} \inf_{h \in \mathcal{H}_b} B_\omega(i, j, h) \geq L_{B,\omega}(q) \right\} \geq 1 - q, \quad (26)$$

$$\mathbb{P} \left\{ |C_\omega(i, j, h)| \leq r_\omega(q) \cdot \|\Delta_{ijh}\|_F \text{ for every } i \neq j \text{ and } h \in \mathcal{H}_b \right\} \geq 1 - q. \quad (27)$$

Finally, we translate these concentration bounds into exact recovery by single linkage. For each true cluster, we select a signal-distance minimum spanning tree and show that, under the separation condition of Theorem 1, every empirical edge in these within-cluster trees

is strictly smaller than every empirical between-cluster distance. The minimum-spanning-tree characterization of single linkage then implies that the true clusters are recovered when the dendrogram is cut at K clusters. We use the following result on single-linkage clustering.

Lemma 8. *Suppose that, for each true cluster $\mathcal{C}_a, 1 \leq a \leq K$, there exists a spanning tree $\mathcal{T}_a \in \mathbb{T}(\mathcal{C}_a)$ such that $\max_{1 \leq a \leq K} \max_{\{i,j\} \in E(\mathcal{T}_a)} \rho(i,j) < \min_{1 \leq a < b \leq K} \min_{i \in \mathcal{C}_a, j \in \mathcal{C}_b} \rho(i,j)$. Then single-linkage hierarchical clustering based on ρ , cut at K clusters, returns $\{\mathcal{C}_1, \dots, \mathcal{C}_K\}$ up to relabeling.*

All these results are proven at the end of this section. We now provide the proof of the theorem using these results. Set $q = \delta/3$, and abbreviate $r = r_\omega(q)$, $U = U_{B,\omega}(q)$, and $L = L_{B,\omega}(q)$. First note that $\underline{\mu}_\omega \leq \bar{\mu}_\omega$, and hence $L \leq \bar{\sigma}^2 p \underline{\mu}_\omega$. Since $c_\omega(q) > 0$, the definition of U then gives $U - L > 0$. Condition (9) consequently implies $\mathcal{B}_\omega - r\sqrt{\mathcal{B}_\omega} > \mathcal{W}_\omega + r\sqrt{\mathcal{W}_\omega} \geq 0$, so $\mathcal{B}_\omega > r^2$. The function $g_-(x) = x - r\sqrt{x}$ is increasing on $[r^2/4, \infty)$. Since every shifted, oriented signal distance for a between-cluster pair is at least $\mathcal{B}_\omega$, it follows that

$$\min_{1 \leq a < b \leq K} \min_{i \in \mathcal{C}_a, j \in \mathcal{C}_b} \min_{(u,v) \in \{(i,j), (j,i)\}} \min_{h \in \mathcal{H}_b} \left\{ A_\omega(u, v, h) - r\sqrt{A_\omega(u, v, h)} \right\} = \mathcal{B}_\omega - r\sqrt{\mathcal{B}_\omega}. \quad (28)$$

By Lemma 7, with probability at least $1 - 3q = 1 - \delta$, the following bounds hold simultaneously for every ordered pair $i \neq j$ and every $h \in \mathcal{H}_b$:

$$B_\omega(i, j, h) < U, \quad B_\omega(i, j, h) \geq L, \quad |C_\omega(i, j, h)| \leq r\sqrt{A_\omega(i, j, h)}.$$

We concentrate on this event. For each true cluster $\mathcal{C}_a$, choose a spanning tree $\mathcal{T}_a^*$ attaining the minimum in (5). For an edge $\{i, j\} \in E(\mathcal{T}_a^*)$, choose an orientation $(u, v) \in \{(i, j), (j, i)\}$ and a shift $h^* \in \mathcal{H}_b$ attaining $\eta_\omega(i, j)$. Using the decomposition (1) and the strict upper

noise bound gives

$$\rho_\omega(i, j) < U + \eta_\omega(i, j) + r\sqrt{\eta_\omega(i, j)} \leq U + \mathcal{W}_\omega + r\sqrt{\mathcal{W}_\omega}. \quad (29)$$

because $x \mapsto x + r\sqrt{x}$ is increasing. Therefore every chosen within-cluster tree edge is strictly below the rightmost quantity in the above display.

For a between-cluster pair, the same simultaneous event and (28) yield

$$\rho_\omega(i, j) \geq L + \mathcal{B}_\omega - r\sqrt{\mathcal{B}_\omega}. \quad (30)$$

The assumed separation condition is equivalent to $U + \mathcal{W}_\omega + r\sqrt{\mathcal{W}_\omega} \leq L + \mathcal{B}_\omega - r\sqrt{\mathcal{B}_\omega}$. Combining this with (29) and (30) gives strict spanning-tree separation. Lemma 8 therefore implies exact recovery on the simultaneous event, whose probability is at least $1 - \delta$.

6.1 Homoskedastic Case: Proof of (10)

Suppose $\sigma_{r,i} = \sigma$ for every $i = 1, \dots, n$ and $r = 1, \dots, p$. If $p\mu_\omega > 2\sqrt{p\lambda\bar{\mu}_\omega c_\omega(\delta/3)}$, the positive part in $L_{B,\omega}(\delta/3)$ is inactive. Substituting the definitions of $U_{B,\omega}$ and $L_{B,\omega}$ and collecting like terms gives exactly (10).

6.2 Asymptotic Implication: Proof of (3)

In the homoskedastic case, suppose that p is fixed. Take $\delta_T = T^{-2}$ and $q_T = \delta_T/3$, and define $\gamma = c_\omega(q_T) + \log 2$, as in Section 3. Since $N_\omega \leq n(n-1)\tau \leq n^2T$, one has $c_\omega(q_T) = \log(3N_\omega T^2) = O(\log n + \log T)$, and $c_\omega(q_T) \leq \gamma$. The assumption $\omega\gamma/T \rightarrow 0$ implies $\omega/T \rightarrow 0$, so $T/\omega \rightarrow \infty$ and $\tau/T \rightarrow 1$. Consequently, $s_\omega = T/\omega + O(1)$, while $\bar{\mu}_\omega = 2T/\omega + O(1)$ and $\mu_\omega = 2T/\omega + O(1)$, because κ is fixed.

By the definition of γ , the cross-term coefficient is exactly $r_\omega(q_T) = 2\sqrt{2}\lambda^{1/2}\sigma\gamma^{1/2}$.

Also, using $[x]_+ \geq x$, the homoskedastic definitions imply

$$U_{B,\omega}(q_T) - L_{B,\omega}(q_T) \leq \sigma^2 \left[p(\bar{\mu}_\omega - \underline{\mu}_\omega) + 4\sqrt{p\lambda\bar{\mu}_\omega c_\omega(q_T)} + 2\lambda c_\omega(q_T) \right].$$

Since p is fixed, $p(\bar{\mu}_\omega - \underline{\mu}_\omega) = O(1)$. Moreover, $4\sqrt{p\lambda\bar{\mu}_\omega c_\omega(q_T)} = (4\sqrt{2} + o(1))\lambda^{1/2}\sqrt{p\gamma T/\omega}$, while the assumption $\omega\gamma/T \rightarrow 0$ gives $1 + \lambda\gamma = o\{\lambda^{1/2}\sqrt{p\gamma T/\omega}\}$.

Therefore, for all sufficiently large T , $U_{B,\omega}(q_T) - L_{B,\omega}(q_T) \leq 4\sqrt{6}\lambda^{1/2}\sigma^2\sqrt{p\gamma T/\omega}$. Hence condition (3) implies the non-asymptotic separation condition in Theorem 1 with $\delta = \delta_T$. The probability of at least one misclassification is therefore at most T^{-2} .

6.3 Proofs of the Auxiliary Results

Proof of Lemma 5. Consider an active shift $\ell \leq h \leq \tau - \ell$. Let X_W and X_L denote the $\ell \times 2$ regression design matrices whose rows are $(1, s/\ell)$, with $s = -\ell + 1, \dots, 0$ and $s = 1, \dots, \ell$, respectively. Direct calculation gives

$$\det(X_W^\top X_W) = \det(X_L^\top X_L) = \frac{\ell^2 - 1}{12}, \quad \text{tr}(X_W^\top X_W), \text{tr}(X_L^\top X_L) \leq 2\ell.$$

For a positive definite 2×2 matrix, its smallest eigenvalue is at least its determinant divided by its trace. Hence, for $\ell \geq 2$, both smallest eigenvalues are at least $\ell/32$. Consequently, the least-squares coefficient maps $B_W = (X_W^\top X_W)^{-1}X_W^\top$ and $B_L = (X_L^\top X_L)^{-1}X_L^\top$ satisfy $\|B_W\|_{\text{op}}, \|B_L\|_{\text{op}} \leq (32/\ell)^{1/2}$.

Let D_h map the shifted vector $\mathbf{z} = S_h \mathbf{v}$ to $(\hat{\alpha}_L - \hat{\alpha}_W, \hat{\beta}_L - \hat{\beta}_W)^\top$. Since the two regression windows are disjoint at every active shift, $\|D_h\|_{\text{op}}^2 \leq \|B_W\|_{\text{op}}^2 + \|B_L\|_{\text{op}}^2 \leq 64/\ell$. Let E_h be the $\tau \times 2$ matrix whose row at coordinate $h - a$ is $w(a)(1, -a/\ell)$, for $a = 0, \dots, \ell_h - 1$, and

whose remaining rows are zero. The correction in Section 2 is therefore $R_h = E_h D_h$, which immediately gives $\text{rank}(R_h) \leq 2$. Since $0 \leq w(a) \leq 1$, $\ell_h \leq c_0 \ell$, and $a < \ell_h$,

$$\|E_h\|_F^2 \leq \ell_h \left\{ 1 + \left(\frac{\ell_h}{\ell} \right)^2 \right\} \leq c_0 \ell (1 + c_0^2).$$

It follows that $\|R_h\|_F \leq \|E_h\|_F \|D_h\|_{\text{op}} \leq 8c_0^{1/2} (1 + c_0^2)^{1/2} = \kappa$. Finally, S_h is orthogonal, so $\|C_h\|_{\text{op}} \leq \|I_\tau + R_h\|_{\text{op}} \leq 1 + \|R_h\|_F \leq 1 + \kappa$. If $h < \ell$ or $h > \tau - \ell$, then $C_h = S_h$, so the result holds with $R_h = 0$.

Applying this same map separately to all p coordinates gives the aggregate operator $I_p \otimes C_h$. The Kronecker-product identities: $\|I_p \otimes C_h\|_{\text{op}} = \|C_h\|_{\text{op}}$, $\text{rank}(I_p \otimes R_h) = p \text{rank}(R_h)$, $\|I_p \otimes R_h\|_F = \sqrt{p} \|R_h\|_F$, imply the stated result.

Proof of Lemma 6. For coordinate r , recall that $\Psi_{\omega,ijh}^{(r)} = \sigma_{r,i}^2 \Sigma_\omega + \sigma_{r,j}^2 C_h \Sigma_\omega C_h^\top$. We write $C_h = (I_\tau + R_h) S_h$ and $M_h = S_h \Sigma_\omega S_h^\top$. Then M_h is positive semidefinite, $\text{tr}(M_h) = s_\omega$, and $\|M_h\|_{\text{op}} \leq 1$. Expanding the corrected trace gives

$$\text{tr}(C_h \Sigma_\omega C_h^\top) = s_\omega + 2 \text{tr}(R_h M_h) + \text{tr}(R_h M_h R_h^\top).$$

Lemma 5 and $\text{rank}(R_h) \leq 2$ imply $\|R_h\|_* \leq \sqrt{2} \|R_h\|_F \leq \sqrt{2} \kappa$. Therefore $|\text{tr}(R_h M_h)| \leq \sqrt{2} \kappa$, while $0 \leq \text{tr}(R_h M_h R_h^\top) \leq \kappa^2$. It follows that

$$\left(s_\omega - 2\sqrt{2}\kappa \right)_+ \leq \text{tr}(C_h \Sigma_\omega C_h^\top) \leq s_\omega + 2\sqrt{2}\kappa + \kappa^2.$$

Combining these inequalities with the variance bounds gives, for every coordinate r , $\sigma^2 \mu_\omega \leq \text{tr}(\Psi_{\omega,ijh}^{(r)}) \leq \bar{\sigma}^2 \bar{\mu}_\omega$. Moreover, $\|C_h \Sigma_\omega C_h^\top\|_{\text{op}} \leq \|C_h\|_{\text{op}}^2 \|\Sigma_\omega\|_{\text{op}} \leq (1 + \kappa)^2$, and hence $\|\Psi_{\omega,ijh}^{(r)}\|_{\text{op}} \leq \bar{\sigma}^2 \lambda$. Finally, because $\Psi_{\omega,ijh}^{(r)}$ is positive semidefinite, we have $\|\Psi_{\omega,ijh}^{(r)}\|_F^2 \leq \|\Psi_{\omega,ijh}^{(r)}\|_{\text{op}} \text{tr}(\Psi_{\omega,ijh}^{(r)}) \leq \bar{\sigma}^4 \lambda \bar{\mu}_\omega$. Next, as aggregate covariance $\Psi_{\omega,ijh}$ is block diagonal, it

implies

$$\text{tr}(\Psi_{\omega,ijh}) = \sum_{r=1}^p \text{tr}(\Psi_{\omega,ijh}^{(r)}), \quad \|\Psi_{\omega,ijh}\|_{\text{op}} = \max_{1 \leq r \leq p} \|\Psi_{\omega,ijh}^{(r)}\|_{\text{op}}, \quad \|\Psi_{\omega,ijh}\|_F^2 = \sum_{r=1}^p \|\Psi_{\omega,ijh}^{(r)}\|_F^2.$$

Summing the trace and squared Frobenius bounds, and taking the maximum of the operator-norm bounds, proves Lemma 6.

Proof of Lemma 7. We use the following standard Gaussian quadratic-form inequalities to prove this result. If $\mathbf{Z} \sim N(\mathbf{0}, M)$, with M positive semidefinite, then, for every $x > 0$,

$$\mathbb{P}\left(\|\mathbf{Z}\|_2^2 \geq \text{tr}(M) + 2\sqrt{\|M\|_F^2 x} + 2\|M\|_{\text{op}}x\right) \leq e^{-x}, \quad (31)$$

$$\mathbb{P}\left(\|\mathbf{Z}\|_2^2 \leq \text{tr}(M) - 2\sqrt{\|M\|_F^2 x}\right) \leq e^{-x}. \quad (32)$$

Fix $i \neq j$ and $h \in \mathcal{H}_b$. Set $\mathbf{Z}_{ijh} = \text{vec}(\boldsymbol{\eta}_{ijh})$. Then $\mathbf{Z}_{ijh} \sim N(\mathbf{0}, \Psi_{\omega,ijh})$ and $B_{\omega}(i, j, h) = \|\boldsymbol{\eta}_{ijh}\|_F^2 = \|\mathbf{Z}_{ijh}\|_2^2$. Now, applying (31) to $\mathbf{Z}_{ijh}$, followed by Lemma 6, shows that, for every $x > 0$,

$$\mathbb{P}\left\{B_{\omega}(i, j, h) \geq \bar{\sigma}^2 \left[p\bar{\mu}_{\omega} + 2\sqrt{p\lambda\bar{\mu}_{\omega}x} + 2\lambda x\right]\right\} \leq e^{-x}.$$

Taking $x = c_{\omega}(q) = \log(N_{\omega}/q)$ and applying a union bound over the $N_{\omega} = n(n-1)|\mathcal{H}_b|$ ordered pair-shift triples proves (25). Similarly, (32) gives

$$\mathbb{P}\left\{B_{\omega}(i, j, h) \leq \bar{\sigma}^2 p\bar{\mu}_{\omega} - 2\bar{\sigma}^2 \sqrt{p\lambda\bar{\mu}_{\omega}x}\right\} \leq e^{-x}.$$

The same union bound, together with the nonnegativity of $B_{\omega}(i, j, h)$, proves (26).

Next, we prove the signal-adaptive control on the cross term. Conditional on the

deterministic signals,

$$C_\omega(i, j, h) = 2\langle \Delta_{ijh}, \boldsymbol{\eta}_{ijh} \rangle_F = 2 \text{vec}(\Delta_{ijh})^\top \text{vec}(\boldsymbol{\eta}_{ijh})$$

is centered Gaussian. By Lemma 6, its variance is at most $4\bar{\sigma}^2\lambda \|\text{vec}(\Delta_{ijh})\|_2^2 = 4\bar{\sigma}^2\lambda \|\Delta_{ijh}\|_F^2$.

Thus, for any $z > 0$, the two-sided Gaussian tail inequality gives

$$\mathbb{P}\left\{|C_\omega(i, j, h)| > 2\sqrt{2}\bar{\sigma}\lambda^{1/2}z^{1/2}\|\Delta_{ijh}\|_F\right\} \leq 2e^{-z}.$$

If $\Delta_{ijh} = \mathbf{0}$, the cross term is identically zero and the same conclusion holds. Taking $z = c_\omega(q) + \log 2 = \log(2N_\omega/q)$ and applying a union bound over all ordered pair-shift triples proves the lemma, because the multiplier is exactly $r_\omega(q)$ from (7).

Proof of Lemma 8. Let the two sides of the displayed inequality be D_{tree} and D_{out} , and choose $t \in (D_{\text{tree}}, D_{\text{out}})$. Form a graph on the observations by connecting i and j whenever $\rho(i, j) \leq t$. Every true cluster is connected because its chosen spanning tree is present, while no edge joins two different true clusters. The connected components are therefore exactly the true clusters. Since single linkage returns the connected components of this threshold graph, cutting its dendrogram at K clusters recovers the true partition.

7 Proof of Lemma 1

Let $\mathcal{E}_\omega = \{\min_{\pi \in \Pi_K} \sum_{i=1}^n \mathbf{1}\{\widehat{I}_i \neq \pi(I_i)\} > 0\}$ denote the event of at least one misclassification. Recall that $\Sigma_\omega = P_\tau A_\omega A_\omega^\top P_\tau$. Set $\nu_\omega = \lambda_{\max}(\Sigma_\omega)$ and $\zeta_\omega = \text{tr}(\Sigma_\omega^2)$. For $0 \leq \mathcal{W} \leq \mathcal{B}$, let $V_{\omega,p}(\mathcal{B}, \mathcal{W}) = 8\sigma^2\nu_\omega(\mathcal{B} + \mathcal{W}) + 16p\sigma^4\zeta_\omega$. Consider $\mathbf{x}_i^{(r)} = \mathbf{g}_i^{(r)} + \sigma\boldsymbol{\varepsilon}_i^{(r)}$, where $r = 1, \dots, p$, $i = 1, \dots, n$, and the vectors $\boldsymbol{\varepsilon}_i^{(r)}$ are independent and distributed as $N(\mathbf{0}, I_T)$. Fix the algorithmic parameters ω, b, ℓ, c_0 , and assume that $\tau = T - \omega + 1 \geq 3$. We next show that

for every $0 \leq \mathcal{W} \leq \mathcal{B} < \infty$ and every $\varepsilon > 0$, there exist $n = 6$ deterministic p -dimensional signals and $K = 3$ true clusters with $\mathcal{B}_\omega = \mathcal{B}$, and $\mathcal{W}_\omega = \mathcal{W}$, such that

$$\mathcal{B} - \mathcal{W} \leq V_{\omega,p}^{1/2}(\mathcal{B}, \mathcal{W}) / (4\sqrt{120}) \text{ and } \mathbb{P}(\mathcal{E}_\omega) \geq 1/1920 - \varepsilon. \quad (33)$$

We provide the proof (33) later on. We next use bounds on ν_ω and ζ_ω from the following lemma and plug those in (33).

Lemma 9. *If $\tau \geq 24\omega$, then $\nu_\omega = \lambda_{\max}(\Sigma_\omega) \geq 1/4$, and $\zeta_\omega = \text{tr}(\Sigma_\omega^2) \geq \tau/(24\omega)$.*

Using these bounds in the definition of $V_{\omega,p}$ we have $V_{\omega,p}(\mathcal{B}, \mathcal{W}) \geq 2\sigma^2(\mathcal{B} + \mathcal{W}) + (2/3)p\sigma^4\tau/\omega$. Let $x = \sigma\sqrt{\mathcal{B} + \mathcal{W}}$ and $y = \sigma^2\sqrt{p\tau/\omega}$. Since $(x + y)^2 \leq 2\{2x^2 + (2/3)y^2\}$, it follows that $\sqrt{V_{\omega,p}(\mathcal{B}, \mathcal{W})} \geq (1/\sqrt{2})\{\sigma\sqrt{\mathcal{B} + \mathcal{W}} + \sigma^2\sqrt{p\tau/\omega}\}$. Let $c_1 = 1/(4\sqrt{240})$. Then, $\mathcal{B} - \mathcal{W} \leq c_1\{\sigma\sqrt{\mathcal{B} + \mathcal{W}} + \sigma^2\sqrt{p\tau\omega^{-1}}\}$ implies that there exists a homoskedastic p -dimensional signal configuration satisfying $\mathcal{B}_\omega = \mathcal{B}$ and $\mathcal{W}_\omega = \mathcal{B}$ for which $\mathbb{P}(\mathcal{E}_\omega) \geq 1/3840$. Finally, because $\omega/T \rightarrow 0$, we have $\tau/T \rightarrow 1$ and $\tau \geq 24\omega$ for all sufficiently large T . The proof of the result therefore follows from the proposition with $c_1 = 1/(4\sqrt{240})$ and $c_2 = 1/3840$. We next provide the proof of (33) and Lemma 9.

Proof of (33). We provide a constructive proof. Let $\mathcal{S}_\tau = \{\mathbf{z} \in \mathbb{R}^\tau : \mathbf{1}_\tau^\top \mathbf{z} = 0\}$ be the centered subspace. Since Σ_ω maps into $\mathcal{S}_\tau$, choose a unit eigenvector $\mathbf{v} \in \mathcal{S}_\tau$ satisfying $\Sigma_\omega \mathbf{v} = \nu_\omega \mathbf{v}$. Choose another unit vector $\mathbf{u} \in \mathcal{S}_\tau$ such that $\mathbf{u}^\top \mathbf{v} = 0$, which is possible because $\tau \geq 3$. Let $\mathbf{e}_1$ denote the first canonical basis vector in $\mathbb{R}^p$. For $\mathbf{a} \in \mathbb{R}^\tau$, write $\mathcal{J}_p(\mathbf{a}) = \mathbf{a}\mathbf{e}_1^\top \in \mathbb{R}^{\tau \times p}$. Thus, $\|\mathcal{J}_p(\mathbf{a})\|_F = \|\mathbf{a}\|_2$, and the coordinatewise correction satisfies $C_h \mathcal{J}_p(\mathbf{a}) = \mathcal{J}_p(C_h \mathbf{a})$. We present the proof of (33) step-wise below.

Step 1: Construction of three two-observation clusters. Let $\mathcal{C}_1 = \{1, 2\}$, $\mathcal{C}_2 = \{3, 4\}$, and $\mathcal{C}_3 = \{5, 6\}$. Choose $M > \sqrt{\mathcal{B}}$, for example $M = \sqrt{\mathcal{B} + 1}$, and define $\mathbf{a}_1 = \mathbf{0}$, $\mathbf{a}_2 = \sqrt{\mathcal{W}}\mathbf{v}$, $\mathbf{a}_3 = M\mathbf{u}$, $\mathbf{a}_4 = M\mathbf{u}$, $\mathbf{a}_5 = M\mathbf{u} + \sqrt{\mathcal{B}}\mathbf{v}$, and $\mathbf{a}_6 = M\mathbf{u} + \sqrt{\mathcal{B}}\mathbf{v}$. Embed these vectors

into p coordinates by setting $\mathbf{A}_i = \mathcal{J}_p(\mathbf{a}_i)$. At the zero shift, $\|\mathbf{A}_1 - \mathbf{A}_2\|_F^2 = \mathcal{W}$, while the corresponding within-cluster distances in $\mathcal{C}_2$ and $\mathcal{C}_3$ are zero. Every pair with one endpoint in $\mathcal{C}_2$ and the other in $\mathcal{C}_3$ has squared Frobenius distance $\mathcal{B}$. All between-cluster distances involving $\mathcal{C}_1$ are strictly larger than $\mathcal{B}$, because $M^2 > \mathcal{B}$ and $\mathbf{u}^\top \mathbf{v} = 0$. Hence the zero-shift geometry has between-cluster separation $\mathcal{B}$ and within-cluster MST radius $\mathcal{W}$.

Step 2: A common anchor that excludes nonidentity shifts. Define $\mathcal{H}_b^\circ = \{h \in \mathcal{H}_b : C_h|_{\mathcal{S}_\tau} \neq I_{\mathcal{S}_\tau}\}$. If $\mathcal{H}_b^\circ = \emptyset$, set $\tilde{\mathbf{f}}_i = \mathbf{A}_i$, and the desired signal geometry already follows. Otherwise, for each $h \in \mathcal{H}_b^\circ$, the set $\mathcal{F}_h = \{\mathbf{q} \in \mathcal{S}_\tau : C_h \mathbf{q} = \mathbf{q}\}$ is a proper linear subspace of $\mathcal{S}_\tau$. Since the shift grid is finite, we may choose a unit vector $\mathbf{q} \in \mathcal{S}_\tau$ such that $C_h \mathbf{q} \neq \mathbf{q}$ for every $h \in \mathcal{H}_b^\circ$.

Let $\mathbf{Q} = \mathcal{J}_p(\mathbf{q})$, and for $L > 0$ define $\tilde{\mathbf{f}}_i^{(L)} = L\mathbf{Q} + \mathbf{A}_i$. At the zero shift, the common anchor cancels. For a nonidentity candidate shift, $\tilde{\mathbf{f}}_i^{(L)} - C_h \tilde{\mathbf{f}}_j^{(L)} = L\mathcal{J}_p(\mathbf{q} - C_h \mathbf{q}) + (\mathbf{A}_i - C_h \mathbf{A}_j)$. Set $d_q = \min_{h \in \mathcal{H}_b^\circ} \|\mathbf{q} - C_h \mathbf{q}\|_2 > 0$ and $R_0 = \max_{i,j,h} \|\mathbf{A}_i - C_h \mathbf{A}_j\|_F < \infty$. Then $\|\tilde{\mathbf{f}}_i^{(L)} - C_h \tilde{\mathbf{f}}_j^{(L)}\|_F \geq Ld_q - R_0$. Thus, for all sufficiently large L , every nonidentity shifted signal distance is larger than every zero-shift signal distance. If $h \notin \mathcal{H}_b^\circ$, then C_h is the identity on each centered coordinate and duplicates the zero-shift distance. Consequently, for sufficiently large L , $\eta_\omega(i, j) = \|\mathbf{A}_i - \mathbf{A}_j\|_F^2$ for every $i \neq j$, which proves $\mathcal{B}_\omega = \mathcal{B}$ and $\mathcal{W}_\omega = \mathcal{B}$. Each column of $\tilde{\mathbf{f}}_i^{(L)}$ lies in $\mathcal{S}_\tau$. Since A_ω has full row rank, $P_\tau A_\omega$ maps onto $\mathcal{S}_\tau$. Applying this surjectivity separately to each coordinate gives a raw signal matrix $\mathbf{g}_i \in \mathbb{R}^{T \times p}$ satisfying $P_\tau A_\omega \mathbf{g}_i = \tilde{\mathbf{f}}_i^{(L)}$. Thus the construction corresponds to valid p -dimensional signals under the model.

Step 3: Two competing empirical edges. Let $\tilde{\boldsymbol{\xi}}_i = (\sigma P_\tau A_\omega \boldsymbol{\epsilon}_i^{(1)}, \dots, \sigma P_\tau A_\omega \boldsymbol{\epsilon}_i^{(p)})$. Then the matrices $\tilde{\boldsymbol{\xi}}_i$ are independent and $\text{vec}(\tilde{\boldsymbol{\xi}}_i) \sim N(\mathbf{0}, \sigma^2 I_p \otimes \Sigma_\omega)$. Consider the zero-shift empirical distances $Q_W = \|\mathbf{v} - \sqrt{\mathcal{W}} \mathbf{v} \mathbf{e}_1^\top + \tilde{\boldsymbol{\xi}}_1 - \tilde{\boldsymbol{\xi}}_2\|_F^2$ and $Q_B = \|\mathbf{v} - \sqrt{\mathcal{B}} \mathbf{v} \mathbf{e}_1^\top + \tilde{\boldsymbol{\xi}}_3 - \tilde{\boldsymbol{\xi}}_5\|_F^2$. Because the two edges involve disjoint observations, Q_B and Q_W are independent. Let

$\mathcal{A}_L = \{\rho_\omega(1, 2) = Q_W\}$. For every nonidentity candidate shift and either orientation, $L^{-2}\|\tilde{\mathbf{f}}_i^{(L)} - C_h\tilde{\mathbf{f}}_j^{(L)} + \tilde{\boldsymbol{\xi}}_i - C_h\tilde{\boldsymbol{\xi}}_j\|_F^2 \xrightarrow{p} \|\mathbf{q} - C_h\mathbf{q}\|_2^2 > 0$, whereas $Q_W = O_p(1)$ as a function of L . Since there are only finitely many candidate shifts, we have $\mathbb{P}(\mathcal{A}_L) \rightarrow 1$ as $L \rightarrow \infty$. Identity shifts give exactly the zero-shift distance. Furthermore, $\rho_\omega(3, 5) \leq Q_B$, because the zero shift is one of the candidate shifts. Hence $\mathcal{A}_L \cap \{Q_B < Q_W\} \subseteq \mathcal{E}_\omega$, and so, we have: $\mathbb{P}(\mathcal{E}_\omega) \geq \mathbb{P}(Q_B < Q_W) - \mathbb{P}(\mathcal{A}_L^c)$. For every $\varepsilon > 0$, L can be chosen sufficiently large that $\mathbb{P}(\mathcal{A}_L^c) \leq \varepsilon$.

Step 4: Mean and variance of the critical comparison. Let $\mathbf{Z}_B = \tilde{\boldsymbol{\xi}}_3 - \tilde{\boldsymbol{\xi}}_5$ and $\mathbf{Z}_W = \tilde{\boldsymbol{\xi}}_1 - \tilde{\boldsymbol{\xi}}_2$. The two matrices are independent, and both vectorizations are distributed as $N(\mathbf{0}, \Gamma_p)$, where $\Gamma_p = 2\sigma^2(I_p \otimes \Sigma_\omega)$. Since their covariances agree, $\mathbb{E}(Q_B - Q_W) = \mathcal{B} - \mathcal{W}$.

For a Gaussian vector $Z \sim N(\mathbf{0}, \Gamma)$, $\text{Var}(\|\delta + Z\|_2^2) = 4\delta^\top \Gamma \delta + 2\text{tr}(\Gamma^2)$. Applying this identity after vectorization, using $\mathbf{v}^\top \Sigma_\omega \mathbf{v} = \nu_\omega$, gives $\text{Var}(Q_B) = 8\sigma^2 \mathcal{B} \nu_\omega + 8p\sigma^4 \zeta_\omega$ and $\text{Var}(Q_W) = 8\sigma^2 \mathcal{W} \nu_\omega + 8p\sigma^4 \zeta_\omega$. Thus, $\text{Var}(Q_B - Q_W) = V_{\omega,p}(\mathcal{B}, \mathcal{W})$. Define $X = Q_B - Q_W - (\mathcal{B} - \mathcal{W})$. Then $\mathbb{E}(X) = 0$ and $\mathbb{E}(X^2) = V_{\omega,p}(\mathcal{B}, \mathcal{W})$. We use the following one-sided lower-tail bound for centered Gaussian quadratic polynomials next. Its proof is provided at the end of this section.

Lemma 10. *Suppose that X is a centered polynomial of degree at most two in a standard Gaussian vector, and let $V = \mathbb{E}(X^2) > 0$. Then, $\mathbb{P}\{X \leq -V^{1/2}/(8\sqrt{30})\} \geq 1/1920$.*

Lemma 10 therefore gives $\mathbb{P}\{X \leq -\sqrt{V_{\omega,p}(\mathcal{B}, \mathcal{W})}/(4\sqrt{120})\} \geq 1/1920$. Now, under the assumption on $\mathcal{B} - \mathcal{W}$ in (33), we have $\mathbb{P}(Q_B < Q_W) \geq 1/1920$. Combining this conclusion with the lower bound on $\mathbb{P}(\mathcal{E}_\omega)$ from Step 3 above provides us with the desired probability bound in (33).

Proof of Lemma 9. These are bounds for the one-coordinate temporal covariance and

are therefore unchanged by the passage from one to p dimensions. Let $Q_\omega = A_\omega A_\omega^\top$. To lower-bound ν_ω , let $r = \lfloor \tau/2 \rfloor$, and define a centered unit vector $\mathbf{z}$ that is constant on the two halves of the domain by $z_t = a$ for $t \leq r$ and $z_t = -c$ for $t > r$, where $a^2 = (\tau - r)/(r\tau)$ and $c^2 = r/[(\tau - r)\tau]$. Then $\min(a^2, c^2) \geq 1/(2\tau)$. Since $\mathbf{z} \in \mathcal{S}_\tau$, $\mathbf{z}^\top \Sigma_\omega \mathbf{z} = \|A_\omega^\top \mathbf{z}\|_2^2$. For every raw-time index $j = \omega, \dots, r$, the corresponding entry of $A_\omega^\top \mathbf{z}$ equals a , and for every $j = r + \omega, \dots, \tau$, it equals $-c$. Hence, when $\tau \geq 4\omega$, $\|A_\omega^\top \mathbf{z}\|_2^2 \geq (\tau - 2\omega + 2) \min(a^2, c^2) \geq 1/4$, so $\nu_\omega \geq 1/4$.

For $\omega \geq 2$, the entries of Q_ω are $(Q_\omega)_{rs} = (\omega - |r - s|)_+/\omega^2$, and therefore $\text{tr}(Q_\omega^2) \geq (2/\omega^4) \sum_{k=1}^{\lfloor \omega/2 \rfloor} (\tau - k)(\omega - k)^2$. When $\tau \geq 24\omega$, this yields $\text{tr}(Q_\omega^2) \geq \tau/(12\omega)$. Let $\lambda_1 \geq \dots \geq \lambda_\tau \geq 0$ be the eigenvalues of Q_ω , and let $\mu_1 \geq \dots \geq \mu_{\tau-1} \geq 0$ be those of its compression to $\mathcal{S}_\tau$. Cauchy interlacing and $\|Q_\omega\|_{\text{op}} \leq 1$ give $\zeta_\omega = \sum_{j=1}^{\tau-1} \mu_j^2 \geq \sum_{j=2}^\tau \lambda_j^2 \geq \text{tr}(Q_\omega^2) - 1 \geq \tau/(24\omega)$. When $\omega = 1$, the same conclusion follows directly from $\Sigma_1 = P_\tau$.

Proof of Lemma 10. There exist a standard Gaussian vector G , a symmetric matrix A , and a vector b such that $X = G^\top A G - \text{tr}(A) + b^\top G$. Write $U = G^\top A G - \text{tr}(A)$ and $R = b^\top G$. The linear and centered quadratic terms are uncorrelated, so $\text{var}(X) = \text{var}(U) + \text{var}(R) = V$. Diagonalizing A gives $E(U^4) = 12\{\text{tr}(A^2)\}^2 + 48\text{tr}(A^4) \leq 60\{\text{tr}(A^2)\}^2$. Since $\text{var}(U) = 2\text{tr}(A^2)$, it follows that $E(U^4) \leq 15\{\text{var}(U)\}^2$. Also, R is Gaussian, and hence $E(R^4) = 3\{\text{var}(R)\}^2$. Using $(x + y)^4 \leq 8(x^4 + y^4)$, we obtain $E(X^4) \leq 120\{\text{var}(U)\}^2 + 24\{\text{var}(R)\}^2 \leq 120V^2$. Interpolation between the L_1 , L_2 , and L_4 norms gives $E|X| \geq \{E(X^2)\}^{3/2}/\{E(X^4)\}^{1/2} \geq V^{1/2}/120^{1/2}$. Let $X_- = \max\{-X, 0\}$. Since $E(X) = 0$, $E(X_-) = E|X|/2 \geq V^{1/2}/(2 \cdot 120^{1/2})$. The Paley–Zygmund inequality therefore yields $\mathbb{P}\{X_- \geq E(X_-)/2\} \geq \{E(X_-)\}^2/\{4E(X_-^2)\} \geq 1/1920$. On this event, $X \leq -E(X_-)/2 \leq -V^{1/2}/(4 \cdot 120^{1/2})$, which proves the result.

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